# The Three-Center Electron-Repulsion Integral Is an Elliptic Bessel Moment: Genus One at the Third Center in Momentum-Space Slater Theory*

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Two-center electron-repulsion integrals over Slater-type orbitals close in elementary functions plus the exponential integral, the logarithm, and Euler's $\gamma$—a genus-zero transcendence class—which is why the two-center problem has been solved since Roothaan and Ruedenberg. The genuine three-center integral has resisted a closed form for seventy years, and the standard verdict is that it is simply hard: reduce it to an infinite series or a one-dimensional quadrature and be done. We give it a structural reading instead. Evaluating the two-electron three-center integral $(XY|XZ)$ over $1s$ orbitals in momentum space—the Fock projection under which Coulomb–Sturmians become hyperspherical harmonics, the setting native to the Averys' generalized-Sturmian program—the three centers enter as three *phases* rather than three foci, so the coordinate obstruction (prolate spheroidal coordinates admit exactly two foci) dissolves. [**MEASURED**] The resulting momentum integral reproduces the exact integral (agreement $1.9\times10^{-14}$ with McMurchie–Davidson on the same Gaussian fits; with the true Slater density, $4.3\times10^{-7}$ against the fitted reference, which itself sits $7\times10^{-8}$ above the Slater value), and its angular part collapses to a single spherical Bessel kernel. [**MEASURED / classical**] What remains—the radial content—is a two-scale Bessel moment whose period is a *complete elliptic integral* (verified to 31 digits), living on the genus-one curve $y^2=(c_1k^2+1)(c_2k^2+1)$; it degenerates to the genus-zero two-center case exactly on the diagonal $c_1=c_2$. [**SYMBOLIC + MEASURED**] The one-mass slice is the Laplace transform of the elliptic differential—an elliptic deformation of the Bessel $K_0$, reducing to $K_0(D)$ when one scale decouples ($\rho\to0$) and to the complete elliptic integral $K$ at zero separation—and satisfies an explicit fourth-order Picard–Fuchs equation with polynomial coefficients (residual $\sim10^{-24}$), whose characteristic exponents are the four branch points of the curve. [**OBSERVATION**] This places a molecular integral inside the modern theory of Bessel moments and two-mass elliptic Feynman integrals; a targeted literature search finds that connection unmade in either field. [**OPEN**] We do *not* give the finite closed form, and we show the natural route to it is obstructed: the elliptic-dilogarithm (unequal-mass sunrise) mechanism does not engage, because the modulus Picard–Fuchs source is not an elementary subtopology but lies in the integral's own Gauss–Manin $D$-module—the branch-point boundary that would supply the sunrise's tadpole vanishes identically. [**OBSERVATION**] The object is therefore an apparently unclaimed elliptic transcendent one level *above* the elliptic-dilogarithm tower, a period of a rank-four irregular connection. [**SYMBOLIC + MEASURED**] That connection is *irreducible*—it is the Fourier–Laplace transform of an irreducible connection, and its monodromy admits no invariant coordinate subspace—so no closed form arises through factorization; [**OPEN**] whether the irreducible object admits a transcendental closed form we leave open. The message is not a solution but a relocation: the third center does not make the radial integral messier within the two-center class, it raises the transcendence one genus, past the reach of the sunrise's elliptic-dilogarithm mechanism. [**MEASURED**] Two pre-registered extensions (Sec. VIII) probe the rung variable: a fourth distinct center adds no genus, consistent with the class counting the distinct two-center densities a kernel couples and not the nuclei, while a three-electron chain over two-center densities with two Coulomb kernels at one electron lives on a K3 surface whose holomorphic period is a three-scale Bessel moment, a sum of products of two complete elliptic integrals. [**SYMBOLIC + MEASURED**] A third extension finds that the holomorphic period of any tree of linked densities depends only on how many densities it links, and that four linked densities give the first period not built from elliptic ones: its equation is an irreducible fourth-order operator of Calabi–Yau type, with no quadratic relation among its solutions, an entry of the published Calabi–Yau tables; for two to five densities the tree ladder is the banana ladder of Feynman integrals under a Hadamard product with the Legendre family. [**SYMBOLIC + MEASURED**] A fourth finds that a Coulomb ring of three densities does not climb: its operator factors at the fibre into elementary, algebraic and elliptic pieces, one rung below the three-density chain, and at one center it carries a dilogarithm where the chain is rational. With exponential geminals $e^{-\gamma r}$ on two edges, the correlated triangle of explicitly correlated methods sits with the chain on the K3 rung at $\gamma=1$; [**OBSERVATION**] the drop is read as the uniqueness of the Coulomb loop, which factors it into pairwise inverse distances.

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## I. THE TWO-CENTER CLASS, AND THE WALL BEYOND IT

The electron-repulsion integral over Slater-type orbitals on two centers is one of the oldest solved problems in molecular quantum mechanics [40]. In prolate spheroidal coordinates, with the two nuclei at the foci, the Neumann expansion of $1/r_{12}$ terminates against the orbital angular momenta, and every integral reduces to elementary functions together with the exponential integral $E_1$, the logarithm, and Euler's constant $\gamma$. In the transcendence bookkeeping of the Geometric Vacuum program (Paper 18 [70]), these are *embedding-tier* constants—the cost of the continuum two-center geometry rather than content of the discrete graph—and, decisively, they sit at transcendence *weight one*, and no dilogarithm appears, even though the ordered radial integral has the simplex shape that generically produces one. In the direct-Coulomb $(AA|BB)$ class—and there alone—even the prefactor $\pi$ cancels against the Coulomb $4\pi$, leaving a rational combination of $\{1,\, e^{-\lambda R}\}$ whose zeros Lindemann–Weierstrass separates; that is what lets *that block* be *decided* rather than merely counted (Paper 58 [72]). It does not extend to the three genuinely cross classes, which carry $E_1$, ln and $\gamma$ seeds and remain counted—weight one is not the same as decidable.

The genuine three-center integral has no such story. Prolate spheroidal coordinates are built from exactly two foci; a third nucleus has nowhere to sit. Gaussians escape this because two Gaussians on different centers multiply into a single Gaussian on a third point [39]—Slater orbitals have no product theorem—and this, not any Geometric-Vacuum-specific limitation, is the seventy-year-old reason Gaussian bases displaced Slater orbitals for polyatomic quantum chemistry. The modern state of the art for three-center Slater repulsion integrals (Ref. [6]) reduces them to a single infinite expansion or a one-dimensional numerical quadrature; the one celebrated closed form in correlated three-electron theory, Fromm and Hill's analytic evaluation [8], is a *dilogarithm*—a genus-zero object—and belongs to a different integral topology (the one-center, explicitly correlated master integral), which lacks the structure we are about to exhibit.

This paper asks what the three-center integral *is*, transcendentally, and answers: it is an elliptic object. The two-center weight-one closure is a genus-zero fact and it stops at two centers. The third center raises the transcendence one genus, to the elliptic (genus-one) world of Bessel moments and two-mass sunrise Feynman integrals. We establish this constructively, validate every step numerically, place the object in the literature, and mark the finite closed form as open.

*Lineage and scope.* The momentum-space route is not new machinery. Molecular integrals over exponential-type orbitals have long been evaluated by Fourier transformation, molecular orbitals in momentum space going back to Shibuya and Wulfman [5]: the transform of a two-center orbital product is classical [41], its reduction to four-dimensional (Fock) hyperspherical harmonics goes back to Niukkanen [42], and the general one- and two-electron multicenter integrals close, via the Fourier-transform method together with Feynman's identity, to low-dimensional numerical quadrature [43]. Recently the four-center two-electron integral over Slater orbitals has been reduced, by cached Fourier–Bessel coefficients, to one radial quadrature, at about $4 \times 10^5$ integrals per second on one GPU at the contraction stage, with the cache already built, and about $4 \times 10^4$ per second end to end, in a preliminary benchmark [12]; that is the production baseline for any cost claim below. Formulas in elementary functions for three- and four-center Slater integrals have also been given [13], but they drop the cross term in the gradients of two orbitals, on the ground that the cosine between them averages to zero, so they are estimates, not exact closed forms. Momentum-space Coulomb–Sturmians are hyperspherical harmonics under Fock's conformal projection [1], the framework the Averys built into the generalized-Sturmian method [2, 3, 50] and applied to Coulomb–Sturmian repulsion integrals [4]. The closest prior art to the present calculation is Avery's momentum-space route [49]—the same Fock-projection method on the same objects, and the one place the elliptic structure was nearest to visible; a targeted search finds it, like all prior work, carrying the evaluation to expansion and numerics without drawing the genus reading, though this is a search-negative inferred from the published abstracts, not a verified absence. What is new here is the reading: that the residual radial object is a specific elliptic Bessel moment, and hence that a molecular integral is an instance of the elliptic-Feynman-integral class. We work throughout with $1s$ orbitals at a common exponent; the generalization to higher $(n, \ell, m)$ multiplies the integrand by solid harmonics and shifts exponents but does not change the genus, and we return to this in Sec. IX.

## II. THE MOMENTUM-SPACE REDUCTION

Write the chemist integral for two densities $\rho_1 = \chi_X \chi_Y$ and $\rho_2 = \chi_X \chi_Z$, two two-center $1s$ densities that share the center $X$ on different axes. With the Fourier representation of the kernel, $1/|\boldsymbol{r}_1 - \boldsymbol{r}_2| = (2\pi^2)^{-1} \int d^3k\, e^{i\boldsymbol{k}\cdot(\boldsymbol{r}_1 - \boldsymbol{r}_2)}/k^2$, the integral becomes

$$(XY|XZ) = \frac{1}{2\pi^2} \int \frac{d^3k}{k^2}\, \tilde{\rho}_1(\boldsymbol{k})\, \overline{\tilde{\rho}_2(\boldsymbol{k})}, \qquad \tilde{\rho}(\boldsymbol{k}) = \int \rho(\boldsymbol{r})\, e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\, d^3r. \tag{1}$$

Each center now appears only through a phase $e^{i\boldsymbol{k}\cdot\boldsymbol{R}}$: three centers are three phases, not three foci, and there is nothing to diagonalize. Equation (1) is exact; the only content is the density Fourier transform $\tilde{\rho}$ and the $\boldsymbol{k}$ integral.

We verify Eq. (1) two independent ways on the reference geometry $X = (0, 0, 0)$, $Y = (0, 0, 2)$, $Z =$

$(1.5, 0, 0.5)$ (bohr), $\zeta = 1$, for which a McMurchie–Davidson evaluation over 8-Gaussian $1s$ fits gives the reference value $(XY|XZ) = 0.204941722$. That value carries the fit error: as the number of Gaussians grows (6 to 16) the evaluation converges to the Slater value 0.2049416525, $7 \times 10^{-8}$ lower, which an exact expansion in angular channels about the shared center reproduces to $3 \times 10^{-10}$ (`tests/test_pairwise_three_center.py`). **[MEASURED]** Using the closed-form Fourier transform of the Gaussian-fitted density (a product of Gaussians is a Gaussian) in Eq. (1) reproduces the McMurchie–Davidson value on the *same* fits to $1.9 \times 10^{-14}$: the momentum formula, the Fourier and Coulomb conventions, and the $\boldsymbol{k}$-quadrature are correct. Independently, computing the *true* Slater density transform (no Gaussian fit) by a one-parameter reduction of the two-center Yukawa product reproduces that reference to $4.3 \times 10^{-7}$. Both are pinned in the regression suite [74].

*The angular integral closes to a spherical Bessel kernel.* Writing each Slater factor through $e^{-\zeta r} = -\partial_\zeta(e^{-\zeta r}/r)$ and the Feynman parametrization of the resulting Yukawa product, the three phases combine into a *single* phase $e^{i\boldsymbol{k}\cdot\boldsymbol{W}}$ with $\boldsymbol{W}(s,t) = (t-s)\boldsymbol{X} + s\boldsymbol{Y} - t\boldsymbol{Z}$ ($s, t$ the two Feynman parameters), so the angular integral is elementary,

$$\int d\Omega_{\boldsymbol{k}}\, e^{i\boldsymbol{k}\cdot\boldsymbol{W}} = 4\pi\, j_0(k|\boldsymbol{W}|) = 4\pi\, \frac{\sin(k|\boldsymbol{W}|)}{k|\boldsymbol{W}|}. \tag{2}$$

The whole integral reduces to a two-dimensional Feynman integral over a one-dimensional radial ($k$) integral,

$$(XY|XZ) = \frac{8}{\pi}\, \partial_{\zeta_a}\partial_{\zeta_b}\partial_{\zeta_c}\partial_{\zeta_d} \int_0^1 ds \int_0^1 dt \int_0^\infty dk\; j_0(k|\boldsymbol{W}|) \times \frac{e^{-D_1\Delta_1}}{\Delta_1}\, \frac{e^{-D_2\Delta_2}}{\Delta_2}\, \Bigg|_{\zeta=1}, \tag{3}$$

with $\Delta_1 = \sqrt{c_1 k^2 + 1}$, $\Delta_2 = \sqrt{c_2 k^2 + 1}$, $c_1 = s(1-s)$, $c_2 = t(1-t)$, and $D_1 = |X - Y|$, $D_2 = |X - Z|$. The $\partial_\zeta$'s (a bookkeeping device restoring the Slater from the Yukawa) and the $(s, t)$ integrations are elementary operations; the transcendental content is entirely in the radial factor. Equation (3) reproduces the 8-Gaussian reference to $1.8 \times 10^{-6}$ **[MEASURED]**, the residual being the finite-order angular and Feynman quadrature.

## III. THE TRANSCENDENCE JUMP: GENUS ZERO TO GENUS ONE

The radial factor of Eq. (3) carries two dispersion factors $\Delta_1, \Delta_2$ with *different* scales $c_1 \neq c_2$. A single such factor is benign. Under Fock's substitution $k\sqrt{c} = m \sinh\theta$ it linearizes, and **[MEASURED,** $6 \times 10^{-18}$**]**

$$\int_0^\infty \cos(kb)\, \frac{e^{-D\sqrt{c\,k^2+m^2}}}{\sqrt{c\,k^2+m^2}}\, dk = \frac{1}{\sqrt{c}}\, K_0\Big(\tfrac{m}{\sqrt{c}}\sqrt{c\,D^2+b^2}\Big), \tag{4}$$

a Bessel $K_0$: the momentum-space Coulomb–Sturmian object, living on a rational (genus-zero) curve. This is why the two-center engine is weight one.

Two different scales cannot be linearized together. Their product is governed by the algebraic curve

$$y^2 = (c_1 k^2 + 1)(c_2 k^2 + 1), \tag{5}$$

a quartic with four distinct branch points $k = \pm i/\sqrt{c_1}, \pm i/\sqrt{c_2}$ whenever $c_1 \neq c_2$: an *elliptic curve*, genus one. When $c_1 = c_2$ the branch points collide and the quartic becomes a perfect square—rational, genus zero. The cleanest witness is the period. At $D_1 = D_2 = 0$ the radial integral is exactly a complete elliptic integral **[MEASURED, 31 digits; classical]**,

$$\int_0^\infty \frac{dk}{\sqrt{(c_1 k^2 + 1)(c_2 k^2 + 1)}} = \frac{1}{\sqrt{c_{\max}}}\, K(m), \tag{6}$$

with modulus $m = 1 - c_{\min}/c_{\max}$ (nondegenerate, $0 < m < 1$; prefactor verified for both orderings of $c_1, c_2$ in `test_paper59_euclidean_dictionary.py`; in Bessel form it is Eq. (19) of Ref. [21]), while on the diagonal $c_1 = c_2 = c$ it collapses to the elementary $\pi/(2\sqrt{c})$. Since $c_1 = s(1-s)$ and $c_2 = t(1-t)$ range independently over $[0, \frac{1}{4}]$, the full integral (3) is an integral over a *family* of elliptic curves, of modulus $m(s,t) = 1 - c_{\min}/c_{\max}$, degenerate only on the measure-zero locus $s = t$ or $s = 1-t$. Every term generated by the $\partial_\zeta$'s is a meromorphic function on the same curve (5), so the whole integral is a period of this elliptic family.

The structural statement is that the embedding tier of the transcendence taxonomy (Papers 18, 34 [70, 71]) is *genus-graded*: genus zero ($\{E_1, \ln, \gamma\}$) at two centers, genus one (elliptic) at three. The two two-center negatives—no $\pi$, no dilogarithm—are genus-zero properties; the third center does not climb to a higher rung of the polylogarithm weight tower but leaves that tower for the elliptic one. This is the transcendence-level form of "no coordinate system for three foci," and it also explains, after the fact, why an attempt to identify the three-center value against dilogarithm constants was structurally doomed: the constants are in the wrong class.

## IV. THE REDUCTION: AN ELLIPTIC BESSEL MOMENT

The radial factor can be named precisely. Applying Eq. (4) to each factor and Parseval's theorem, the radial core (at fixed Feynman parameters, before the elementary $j_0$ and $\partial_\zeta$ wrapping) is a two-mass Bessel moment **[MEASURED, 30 digits]**,

$$M = \frac{2}{\pi\sqrt{c_1 c_2}} \int_0^\infty db\; K_0\big(a_1\sqrt{p_1^2 + b^2}\big)\, K_0\big(a_2\sqrt{p_2^2 + b^2}\big), \tag{7}$$

with $a_i = 1/\sqrt{c_i}$ and $p_i = D_i\sqrt{c_i}$. This is a shifted-argument, two-scale member of the Bessel-moment family; it is not one of the moments catalogued in closed form by Bailey, Borwein, Broadhurst and Glasser [21], whose two-scale entries (their Eqs. 19 and 30) are the unshifted case $p_1 = p_2 = 0$.

A change of variable $x = \sqrt{c_1 k^2 + 1}$ on the one-mass slice $N(D) \equiv M(D, 0)$ puts it in the cleanest canonical form: the Laplace transform of the holomorphic elliptic differential,

$$N(D) = \frac{1}{\sqrt{c_1}} \int_1^\infty \frac{e^{-Dx}\, dx}{\sqrt{(x^2-1)(\rho\, x^2 + 1 - \rho)}}, \qquad \rho = \frac{c_2}{c_1}. \tag{8}$$

This is a single explicit one-dimensional integral of an elementary integrand. A low-dimensional reduction is the expected output of the Fourier-transform method—the same machinery reduces multicenter Slater integrals to two- or three-dimensional quadrature [43] and three-center integrals to a semi-infinite oscillatory Bessel integral [44]—so what is new is not the reduction itself but the *elliptic reading* of the integrand it produces, which makes the genus-one structure manifest. It is an *elliptic deformation of the Bessel* $K_0$: as $\rho \to 0$ (the one-mass limit, one scale decoupling) $N \to K_0(D)$, and at $D = 0$ it is the complete elliptic integral $K$ of Eq. (6).

Because $1/\sqrt{Q}$, $Q(x) = (x^2-1)(\rho x^2 + 1 - \rho)$, satisfies the first-order relation $Qf' + \frac{1}{2}Q'f = 0$, and the boundary at the branch point $x = 1$ vanishes ($Q(1)^{1/2} = 0$), integration by parts against $e^{-Dx}$ (using $\int_1^\infty x^n e^{-Dx} f\, dx = (-1)^n L^{(n)}(D)$, $L \equiv \sqrt{c_1}\, N$) yields an exact fourth-order Picard–Fuchs equation with polynomial coefficients **[SYMBOLIC PROOF; MEASURED residual** $\sim 10^{-24}$**]**,

$$D\rho\, L'''' + 2\rho\, L''' + D(1-2\rho)L'' + (1-2\rho)L' - D(1-\rho)L = 0. \tag{9}$$

Its characteristic polynomial factors as $\rho(s^2-1)\big(s^2 + (1-\rho)/\rho\big)$, whose roots $s = \pm 1$ and $s = \pm i\sqrt{(1-\rho)/\rho}$ are exactly the four branch points of the curve (5): the $K_0/I_0$ sector from the $(x^2-1)$ pair, the oscillatory $J_0/Y_0$ sector from the other. (A differential-equation method for molecular two-electron integrals is itself established: Pachucki [48] evaluates the *two*-center integral from a second-order equation in the internuclear distance, closing it in elementary functions; the genus jump appears here as the step from that second-order, genus-zero equation to the fourth-order Picard–Fuchs equation of an elliptic curve.) A subordination identity casts the same object as a two-scale sunrise integral, $\propto \iint ds_1 ds_2\, (s_1 s_2)^{-1/2} e^{-D_1^2/4s_1 - s_1 - D_2^2/4s_2 - s_2} / \sqrt{s_1 c_1 + s_2 c_2}$, whose second-Symanzik-type $\sqrt{\text{linear form}}$ is the genus-one signature and separates only when $c_1 = c_2$.

One numerical fact pins the object's class beyond the period. $N(D)$ is *non-analytic in* $D$ *at* $D = 0$: it carries a $D \ln D$ term **[MEASURED]**, so it is not a pure elliptic integral ($K/E$ are analytic in their parameters). The elliptic $D = 0$ period together with the $D \ln D$ non-analyticity are the two fingerprints of an *elliptic* $\oplus$ *logarithmic* object—the inhomogeneous ("regulator") term of the Picard–Fuchs solution associated with the elliptic dilogarithm [28, 29, 51]—a correspondence whose reach we test, and bound, in Sec. V.

**[SYMBOLIC + MEASURED]** *The kinematics.* Equation (4) is the frequency representation of the two-dimensional Euclidean propagator: with $G_a(\tau, x) = (2\pi)^{-1} K_0(a\sqrt{\tau^2 + x^2})$ the Green's function of $-\Delta + a^2$ on $\mathbb{R}^2$, its left-hand side is exactly $(2\pi/\sqrt{c})\, G_a(p, b)$ with mass $a = \zeta/\sqrt{c}$, Euclidean time $p = D\sqrt{c}$ and space $b$, and the fibre (7) is exactly the two-propagator correlator $M = 4\pi a_1 a_2 \int_{\mathbb{R}} db\, G_{a_1}(p_1, b)\, G_{a_2}(p_2, b)$: two Euclidean packets of unequal mass, evolved from a point source for Euclidean times $p_1, p_2$ and overlapped on one spatial slice **[MEASURED,** $10^{-30}$ **and** $10^{-31}$ **at five parameter points each;** `test_paper59_euclidean_dictionary.py`**]**. Three things become visible. First, the Feynman parameters are Källén–Lehmann spectral variables: $c = s(1-s)$ forces $a = \zeta/\sqrt{c} \geq 2\zeta$, so each two-centre density is dispersed over its own two-particle cut and the elliptic modulus $\rho = c_2/c_1 = (a_1/a_2)^2$ is the squared ratio of the two spectral masses—the genus jump is a mass-splitting phenomenon, and the three cusps are equal masses ($\rho = 1$) and one mass at the heavy end of its cut ($\rho \to 0, \infty$). Second, the exponential twist is $a_i p_i = \zeta D_i$, *independent of the Feynman parameter*: the family is swept at fixed mass×time, and the physical point is one Compton wavelength of Euclidean-time displacement, the $u = D$ endpoint of the source-time superposition $e^{-D\Delta}\Delta^{-n} = \frac{1}{(n-1)!}\int_D^\infty (u-D)^{n-1} e^{-u\Delta}\, du$ that converts the Yukawa into the Slater. Third, $j_0(k|\boldsymbol{W}|)$ is a uniform smearing of the shared spatial coordinate over the composite separation, so the third centre enters *only* through space while the bond lengths set the two Euclidean-time offsets **[MEASURED,** $1.8 \times 10^{-10}$ **for the full position-space fibre]**.

## V. THE ELLIPTIC-DILOGARITHM ROUTE IS OBSTRUCTED

The unequal-mass sunrise is closed by variation of parameters: a second-order Picard–Fuchs operator in the external variable annihilates the two periods, its inhomogeneity is an elementary subtopology (a tadpole), and the particular solution—an iterated integral of the periods against that elementary source—is the elliptic dilogarithm [28, 29]. (The full unequal-mass sunrise has since been closed to all orders in $\varepsilon$ as iterated integrals on $\overline{\mathcal{M}}_{1,3}$ by Bogner, Müller-Stach and Weinzierl [30]—a *Fuchsian* object; the point below is that $N(D)$ is its irregular Laplace dual and so lies outside that technology.) Running the same machinery on $N(D)$ is the natural attempt. It does not close, and the reason is structural.

The curve $y^2 = Q(x, \rho)$ of Eq. (5) moves with the mod-

ulus $\rho$, not with the Laplace variable $D$. Accordingly the two periods $K(\rho), K(1-\rho)$ are the homogeneous solutions of the second-order modulus operator [**MEASURED: both annihilated to** $10^{-25}$**;** $L(0,\rho) = K(1-\rho)$ **to** 19 **digits**]

$$\mathcal{M}_\rho = \rho(1-\rho)\,\partial_\rho^2 + (1-2\rho)\,\partial_\rho - \tfrac{1}{4}, \tag{10}$$

and they are *not* solutions of the fourth-order operator in $D$: a constant gives $\mathcal{O}[1] = -D(1-\rho) \neq 0$, and the single Bessel functions of both sectors, $K_0(D), I_0(D)$ and $J_0(\omega D), Y_0(\omega D)$ with $\omega = \sqrt{(1-\rho)/\rho}$, are likewise not $D$-solutions [**MEASURED**] (they are coupled, which is why Eq. (9) does not factor into elementary Bessel operators). The elliptic-dilogarithm construction lives on the periods, so it must be run in the modulus $\rho$.

There it meets the obstruction. Acting with $\mathcal{M}_\rho$ under the integral sign is an exact total derivative [**SYMBOLIC PROOF; exact residual** 0],

$$\mathcal{M}_\rho\big[Q^{-1/2}\big] = \frac{d}{dx}\Big[\, g(x)\, Q^{-1/2}\Big], \quad g(x) = \frac{-x(x^2-1)}{4(\rho x^2 + 1 - \rho)}, \tag{11}$$

so for $D \neq 0$ the modulus source is $S(D,\rho) \equiv \mathcal{M}_\rho[L] = D\int_1^\infty e^{-Dx} g\, Q^{-1/2} dx$. Because $g$ *vanishes at the branch point* $x = 1$, differentiating in the modulus produces no branch-point boundary term: the analogue of the sunrise's pinched propagator—the elementary tadpole that would be the inhomogeneity—is absent, and the source stays on the elliptic curve. Quantitatively [**MEASURED**], $S$ closes exactly and finitely only inside the integral's own Gauss–Manin $D$-module,

$$S(D,\rho) = \sum_{k=0}^{3} P_k(D,\rho)\,\partial_D^k L, \qquad \deg_D P_k = 2, \tag{12}$$

with held-out residual saturating flat under increase of the fit degree (below $10^{-25}$ in the regression witness, $5\times10^{-44}$ at driver precision; numerically an exact finite identity); whereas the genuine variation-of-parameters shape—period level $\{L, \partial_D L\}$ plus an elementary Bessel subtopology $\{K_0, K_1\}(D)$, $\{J_0, Y_0\}(\omega D)$—only *approximates* it (residual $2\times10^{-14}$ falling geometrically to $3\times10^{-30}$ across fit degrees, never saturating). The inhomogeneity is in-module, not a simpler subtopology, and variation of parameters against the periods is therefore circular.

[**OBSERVATION**] $N(D)$ is thus one transcendence level above the sunrise—above the *fully solved* sunrise, not merely its leading elliptic dilogarithm: the all-orders unequal-mass sunrise is now closed as iterated integrals on $\overline{\mathcal{M}}_{1,3}$ [30], a period of a *Fuchsian* connection, whereas $N(D)$ is not. It is not an elliptic dilogarithm of elementary or period-level arguments; it is a period of the rank-four irregular connection defined by Eq. (9), for which the sunrise mechanism supplies no reduction to the elliptic-polylogarithm tower [67]. [**SYMBOLIC**] The irregularity is explicit at the operator level: at $D = \infty$ the operator of Eq. (9) admits four formal exponential solutions $e^{\lambda D}$ with $\lambda \in \{\pm1, \pm i\sqrt{(1-\rho)/\rho}\}$—its leading Newton edge is a full-width slope-one segment, so it is irregular of Poincaré rank one, the four $\lambda$ being exactly the branch points (the $K_0/I_0$ and $J_0/Y_0$ Bessel sectors), whereas the sunrise/Legendre period operator is regular singular at infinity ($\lambda = 0$ only). The two operators therefore sit on opposite sides of the Laplace transform, and the Fuchsian elliptic-polylogarithm machinery of [30] cannot reach $N(D)$. Whether $N(D)$ admits a closed form of any kind is a differential-Galois question—whether the fourth-order operator factors over the differential field generated by the periods $K(\rho), K(1-\rho)$—settled below, negatively, at the reducibility level (the operator does not factor); what remains open is only whether a *transcendental* closed form exists at the irreducible level.

[**MEASURED**] The same irregularity is visible in $N(D)$ itself, as a *resurgent* structure. Expanded about the Bessel point $\rho = 0$ ($N \to K_0(D)$), the one-mass slice has the explicit elementary term series

$$g(1) = 0, \qquad N(D) = \frac{1}{\sqrt{c_1}} \sum_{n\geq 0} (-1)^n \frac{\big((2n-1)!!\big)^2}{2^n\, n!}\, \rho^n\, \frac{K_n(D)}{D^n}, \tag{13}$$

each coefficient a rational multiple of an *integer-order* Bessel function—the transcendence does not escalate—but the series is *asymptotic*, diverging factorially (optimal truncation at $n \sim$ 1–3, a few digits, then unbounded). Equivalently the $D \to \infty$ expansion $N(D) \sim e^{-D}\sum_k b_k D^{-(k+1/2)}$ is Gevrey-one, and its Borel transform is the algebraic function $\psi(\zeta) = [(\zeta+2)(\rho(1+\zeta)^2 + 1 - \rho)]^{-1/2}$ on the very curve (5), singular exactly at the three non-dominant branch points $\zeta \in \{-2, -1 \pm i\sqrt{(1-\rho)/\rho}\}$—the Stokes actions linking the $e^{-D}$ sector to the $e^{+D}$ and $e^{\pm i\sqrt{(1-\rho)/\rho}\,D}$ ones. A Borel–Padé resummation confirms these locations (at $\rho = \frac{1}{5}$: $\zeta = -2$ and $-1 \pm 2i$, recovered to $\sim 1\%$). So the residual transcendence of $N(D)$ is not exotic: it is the resurgence of an elementary Bessel series, its Stokes data pinned to the branch points of the elliptic curve.

[**MEASURED**] The resurgence, moreover, *closes with algebraic Stokes data.* The Borel transform $\psi(\zeta)$ carries only square-root singularities, so its amplitude at each is algebraic: $a_* = 1$ at $\zeta = -2$, and $a_*^2 = -\frac{1}{2} \mp \frac{i}{2}\sqrt{\rho/(1-\rho)}$ at $\zeta = -1 \pm i\sqrt{(1-\rho)/\rho}$ (matched to the closed-form residue to $10^{-37}$, and confirmed to be the large-order growth constants by singularity analysis; the dominant Stokes ray switches from the real to the complex pair at $\rho = \frac{1}{4}$). These amplitudes *are* the Stokes constants connecting the four exponential sectors, so the entire trans-series of $N(D)$ over $\{e^{\pm D}, e^{\pm i\sqrt{(1-\rho)/\rho}\,D}\}$ has *algebraic* connection data. The one genuinely elliptic ingredient enters only at the boundary: at $D = 0$ the open Laplace thimbles close into the cycles of the curve and $\sqrt{c_1}\,N(0) = K(1-\rho)$ exactly (14 digits). So the closed form at the irregular level is explicit: $N(D)$ is the Borel–Laplace transform of the algebraic differential $\psi$, with algebraic Stokes data and an elliptic-period

$K(1-\rho)$ normalization—as far as a genuinely irregular, genus-one object reduces. What remains transcendental is the elliptic period at the boundary and the standard Borel–Laplace resummation, not any new constant.

[**OBSERVATION**] Structurally this is the genus-one analogue of the classical resurgence of the modified Bessel function $K_0$—the archetypal rank-one irregular equation, whose large-argument series has an algebraic Borel transform and elementary Stokes data [31, 32]. The natural home for the algebraic-Stokes-versus-transcendental-period dichotomy seen here is Sabbah's theory of Stokes structures for Fourier–Laplace transforms of algebraic connections [33], in which the Stokes data is local at the irregular point (hence algebraic) while transcendental periods enter through the regular points (hence $K(1-\rho)$ at $D=0$). The closest worked instance is the rank-two, genus-*zero* Bessel/Kloosterman connection treated via irregular Hodge theory [37]; and the only resurgence analysis of the sunrise/banana family we found operates in the modular parameter $q=e^{2\pi i\tau}$ (the $|q|\to 1$ limit), not the Laplace-dual $D$ [34]. The present genus-one, Laplace-dual reading appears unmade; an absence from a targeted search, not a proof.

*A concrete route from the solved dual.* Because $N(D)$ is the Laplace transform of the sunrise elliptic differential and the unequal-mass sunrise is now known in full closed form [30], one concrete—if not obviously tractable—route to $N(D)$ is to Laplace-transform that solution directly, an operation that necessarily leaves the elliptic-polylogarithm class for the Bessel-moment/resurgence one, consistent with the irregular character just established. We have not carried this out.

*The rank-four label is not vacuous: a base-field reducibility scan.* That $N(D)$ is a period of a genuinely rank-four object—rather than one secretly carrying lower-rank content—is itself a reducibility statement, and it can be tested directly. Two structural facts constrain Eq. (9). First, the operator is *formally self-adjoint*, $L^*=L$ [**SYMBOLIC PROOF**], so its differential Galois group lies in the symplectic group $\mathrm{Sp}_4(\mathbb{C})$—the signature of a polarized period. Second, its indicial exponents at the regular singular point $D=0$ are $\{0,1,1,2\}$ [**SYMBOLIC PROOF**]; the *double* exponent at 1 is the local-monodromy source of the $D\ln D$ nonanalyticity noted above. A search over the base field $\mathbb{Q}(\rho)(D)$ then returns *no* factor of order one or two—hence, by self-adjointness, none of order three—with polar locus at the true singularities $\{0,\infty\}$ [**SYMBOLIC, positive-control validated**]: there is no hyperexponential solution $e^{\lambda D}Q(D)$ for any $\lambda\in\{\pm1,\pm i\omega,0\}$, and the only two order-two factorizations the symbol $(s^2-1)(s^2+\omega^2)$ admits over $\mathbb{Q}(\rho)$—the two Bessel sectors—both fail. The two natural elementary structures are excluded by the exponents alone: $L$ is neither the symmetric product of its Bessel sectors (which would carry exponents $\{0,0,0,0\}$) nor a symmetric cube. This is strong evidence that Eq. (9) is irreducible over $\mathbb{Q}(\rho)(D)$, and hence that the rank-four label is earned. The eigenring sharpens it to a single residual target. The endomorphism algebra of the solution space is trivial, $E(L)=\mathbb{C}$ [**computed at $\rho=1/5$; robust across pole windows**], so $L$ is indecomposable; and the symplectic form pairs the exponential solution $y_\lambda$ at $D=\infty$ with $y_{-\lambda}$. A non-degenerate invariant subspace would split $V=W\oplus W^\perp$ and enlarge the eigenring, so every invariant subspace is *isotropic*; the exponential torus $T=(\mathbb{C}^*)^2$ confines invariant subspaces to spans of the four lines $\{y_{\pm1},y_{\pm i\omega}\}$, whose two symplectic pairs (the Bessel sectors) are non-degenerate. A one-dimensional isotropic invariant subspace is a hyperexponential solution, already excluded. Hence $L$ is reducible over $\mathbb{C}(D)$ *iff* one of the four "mixed" Lagrangian planes $\mathrm{span}\{y_{\pm1},y_{\pm i\omega}\}$ is invariant—equivalently, iff the Stokes matrices at $D=\infty$ [33] fail to couple the real exponential pair $\{\pm1\}$ to the imaginary pair $\{\pm i\omega\}$. This is decidable, and it is settled twice over—once in principle, once explicitly.

*In principle.* The reduction is an instance of the Fourier–Laplace transform of $D$-modules, where irreducibility is automatic. Let $\mathcal{L}$ be the rank-one meromorphic connection on $\mathbb{P}^1$ whose flat sections are $Q^{-1/2}$: it is regular singular, with local monodromy $-1$ at each of the four branch points $x\in\{\pm1,\pm i\omega\}$ and trivial monodromy at $\infty$ (there $Q^{-1/2}\sim x^{-2}$). Having rank one, $\mathcal{L}$ is irreducible, so its minimal (intermediate) extension $j_{!*}\mathcal{L}$ to $\mathbb{A}^1$ is a *simple* holonomic $D$-module. (Because every finite local monodromy is $-1\neq 1$ and hence has no invariant, $j_!\mathcal{L}=j_{!*}\mathcal{L}=j_*\mathcal{L}$, so "the $D$-module generated by $Q^{-1/2}$" is unambiguous.) The Laplace kernel $e^{-Dx}$ makes the radial integral the Fourier–Laplace transform $\mathrm{FL}(j_{!*}\mathcal{L})$, and FL is an exact auto-equivalence of the category of holonomic $D$-modules on the affine line (with its dual coordinate) [38]; an equivalence sends simple objects to simple objects, so $\mathrm{FL}(j_{!*}\mathcal{L})$ is simple. Its generic rank is the total drop of $\mathcal{L}$—one unit at each branch point, where the monodromy $-1$ leaves no invariant, and none at the regular point $\infty$, hence *four*, matching the order of Eq. (9), of which $N(D)$ is the defining (cyclic) solution. A simple holonomic $D$-module of generic rank four restricts to an irreducible connection over the function field, so **the operator of Eq. (9) is irreducible** over $\mathbb{C}(D)$, hence over $\mathbb{Q}(\rho)(D)$. Simplicity is a property of the connection at the generic point, so it forbids a factor of *any* order—apparent singularities and all; the monodromy computation next verifies it by hand.

*Explicitly.* The same conclusion is exhibited, and the mechanism made visible, by the monodromy. The four exponentials are the branch points $x\in\{\pm1,\pm i\omega\}$ of $y^2=Q$ in the Borel (Laplace-dual) plane, and the monodromy of $L$ around $D=0$, computed in the exponential-line basis of Lefschetz-thimble solutions $\int e^{-Dx}\,dx/\sqrt{Q}$, is [**MEASURED; exact-integer certificate**] the integer unipotent matrix

$$M_0=\begin{pmatrix}-1&2&2&2\\-2&3&2&2\\-2&2&3&2\\2&-2&-2&-1\end{pmatrix},$$

the *same* for $\rho = \frac{1}{5}$ and $\rho = \frac{1}{3}$ (a topological invariant of the branch-point configuration), with a single $2 \times 2$ Jordan block—the lone $D \ln D$ log—and *no* invariant coordinate subspace. The Stokes matrices therefore *do* couple the real and imaginary exponential pairs, no mixed plane survives, and the reducibility criterion above returns *irreducible*—confirming the structural argument concretely. **[SYMBOLIC + MEASURED]** Consequently $N(D)$ is a period of an *irreducible* rank-four connection: it admits no closed form through any factorization of $L$, and the sole remaining route to a closed form is a genuinely transcendental one at the full rank-four level—the elliptic-polylogarithm / $\Gamma(2)$ multiple-modular-value structure developed in the companion (Paper 61 [73]). The differential-Galois *reducibility* question is thereby settled (negatively: $L$ is irreducible); the transcendental closed form is the one problem that remains open [74].

## VI. PLACEMENT IN THE LITERATURE

The object of Eqs. (7)–(9) sits inside a fully developed body of mathematics, applied so far only to problems far from chemistry. Two-mass elliptic Feynman integrals—the sunrise/banana family—are evaluated in terms of elliptic dilogarithms as inhomogeneous solutions of exactly this kind of Picard–Fuchs equation [28, 29, 35]; their periods are the complete elliptic integrals of Eq. (6), and at special (complex-multiplication) scale ratios their values reach critical $L$-values of modular forms [36, 37, 68, 69]. The closest published analog to Eq. (7) is a doubled-argument Bessel moment that Broadhurst evaluated as a product of complete elliptic integrals via a lattice-Green-function contour [23]: the same "two-scale moment $\rightarrow$ product of $K$'s" mechanism we expect at rational $\rho$.

On the chemistry side the treatment is uniformly numerical: three-center Slater repulsion integrals are reduced to infinite series or one-dimensional quadrature [6, 7], and the Averys' momentum-space Coulomb–Sturmian route [4, 49], closest in spirit to ours, likewise carries the evaluation to expansion and numerics with no genus reading our search could find. We could find no paper drawing the bridge in either direction: no chemistry reference names an electron-repulsion integral as a Bessel moment or an elliptic Feynman integral, and no amplitudes reference applies the elliptic machinery to a molecular integral. **[OBSERVATION]** The identification—that the three-center Slater integral, via the Fock projection, is a two-scale Bessel moment on the sunrise elliptic curve—appears to be unmade. This is an absence from a targeted search, not a proof of absence; the residual risk is an obscure single source. **[OBSERVATION]** A second placement datum is arithmetic: the $\mathbb{Z}[i]$ content this family carries (the theta series $\theta_3^2$ of the companion (Paper 61); $G = \beta(2) = L(2, \chi_{-4})$) does not appear in the established Bessel-moment corpus, whose forms sit at levels $3, 6, 8, 15$ [36, 37, 68] and, in the twisted moments of Lim, Tu and Yu [60], which are the moments $c_{n,0}$ of Sec. VIII C, 24 and 48: a targeted search finds no Catalan-constant appearance in any of these, and $\chi_{-4}$ enters $c_{3,0}$ as a twist: the form attached to it is the newform of level 12 twisted by $\chi_{-4}$, whose complex multiplication stays by $\mathbb{Q}(\sqrt{-3})$, not $\mathbb{Q}(i)$ [59]; the nearest other $\chi_{-4}$ appearance is the topological-string sector of the Broadhurst–Dorigoni programme [34]. The family thus occupies an apparently unclaimed arithmetic slot ($\mathbb{Z}[i]$, complex multiplication by $\mathbb{Q}(i)$) in that literature, of a piece with the unmade bridge above. (Reconciliation with the value-level searches of the companion (Paper 61): the conductor-4 arithmetic is a property of the *family* and of the $D \to 0$ shadow; the physical integrated value itself shows no measured conductor-4 content at reachable heights.)

The modular and complex-multiplication structure of this elliptic family, its cosmic-Galois periods, and the Bessel-moment period algebra of the object—the master family, the Wronskian $W_0 D^{-2}$, and the quadratic period relations $B = \pi\,\Omega$ with $\Omega$ the canonical integer symplectic intersection form—are developed in the companion Paper 61 [73], which takes the present reduction as its starting point.

## VII. EXPLICITLY-CORRELATED INTEGRALS ARE NATIVE TO THIS REPRESENTATION

The momentum reduction of Sec. II is *kernel-agnostic*. A two-electron matrix element of *any* operator $f(r_{12})$ between orbital products is, by the convolution theorem,

$$\langle \phi_a \phi_b \,|\, f(r_{12}) \,|\, \phi_c \phi_d \rangle = \frac{1}{(2\pi)^3} \int d^3k\; \widetilde{\rho}_{ac}(\boldsymbol{k})^*\, \widetilde{f}(\boldsymbol{k})\, \widetilde{\rho}_{bd}(\boldsymbol{k}), \qquad \rho_{ac} = \phi_{\!}$$

(14)

the identical object as the electron-repulsion integral with the Coulomb kernel $\widetilde{(1/r_{12})} = 4\pi/k^2$ replaced by $\widetilde{f}$, and the angular closure to the spherical Bessel kernel of Eq. (2) applies unchanged. Explicitly-correlated (F12) methods take $f$ to be a correlation factor; the standard choice is the Slater geminal $f = e^{-\gamma r_{12}}$, whose transform $\widetilde{f}(k) = 8\pi\gamma/(k^2+\gamma^2)^2$ is exact, elementary, and—unlike the Coulomb kernel—*regular* at $k = 0$. In this representation the geminal is therefore a native object: no expansion of $e^{-\gamma r_{12}}$ in Gaussians (the first workaround of Gaussian-basis F12) is required. **[MEASURED]** The momentum form (14) reproduces a direct position-space quadrature to $10^{-14}$, and the Coulomb self-repulsion of a $1s$ density to its exact value $5a/8$ (`test_f12_kernel_swap_and_native_geminal`).

At the level of the curve, whether the integral closes in elementary functions is governed *not* by the operator but by the *density topology*; the class of the integrated object can still move with the kernel's mass and shape (the two-center master integral below, and the correlated triangle of Sec. VIII D). **[SYMBOLIC]** When both transition densities are single-center the radial integrand is rational and the pair integral is genus

zero: e.g. $\int_0^\infty [k^2/(k^2+\gamma^2)^2]\, j_0(kR)\, dk = (\pi/4\gamma)\, e^{-\gamma R}$ (`test_f12_geminal_pair_is_elementary`). A *two*-center transition density, however, contributes a Fock dispersion factor $\sqrt{c\,k^2+1}$ (Sec. IV), and two of them build the very curve of Eq. (5), $y^2 = (c_1k^2+1)(c_2k^2+1)$, whose period is a complete elliptic integral (`test_f12_multicenter_shares_eri_elliptic_genus`). A rational kernel—Coulomb $4\pi/k^2$ or geminal $8\pi\gamma/(k^2+\gamma^2)^2$ alike—is a rational function on this curve, so $\widetilde{f}\,dk/y$ is again an elliptic integral: the genus is *unchanged*, the geminal merely trading an elliptic integral of the first kind (the Coulomb period) for one of the third kind (its poles at $k^2 = -\gamma^2$). [**OBSERVATION**] On every instance tested, the transcendence class is set by the number of two-center transition densities on distinct center pairs—zero or one gives genus zero, two gives genus one, while two densities on the same pair (the two-center exchange class) build the same fibre family yet close in the two-center seed set (Paper 18 [70]), so the fibre genus alone does not decide the class—exactly as for the repulsion integral.

[**OBSERVATION**] The consequence is a *unification*, not an escape. Explicitly-correlated integrals are native to the Fock momentum representation and cost the same as the corresponding repulsion integrals, because here, for the physical integrals tested, the curve of a two-electron integral is built by the densities, not by the operator between them; the kernel's mass can still move the class of the integrated object (the observation closing this section). In particular the elliptic obstruction of this paper—the genuine three-center, two-two-center-density integral—gates explicitly-correlated *evaluation* as well: the corresponding closed form (the kernel changes the kind of period, not the curve) that would evaluate the three-center repulsion integral would evaluate the multicenter F12 correction. What correlation buys is not a cheaper integral but a far more compact basis—a single $r_{12}$ factor recovers the electron–electron cusp that orbital products approach only as $\ell \to \infty$—so the route trades many repulsion integrals for few correlated ones drawn from the same transcendence classes. We do not here treat the many-electron (resolution-of-identity) machinery of a full F12 method, nor a resource comparison with tuned Gaussian F12; the statement is about the two-electron integrals, whose momentum-space transcendence we have mapped.

[**PRIOR ART + SYMBOLIC**] The two-center integral with a Slater geminal shows where this reading needs care. Take the two-electron master integral of a diatomic molecule, with an inverse power and an exponential on each of the five distances. Without a geminal its equation in the bond length has second order and Bessel solutions [48]; with one it has fourth order, and Lesiuk and Moszynski defined its solutions by series and named them hyper-Bessel functions [47]. Their operator is Eq. (9) after a rescaling of the bond length. Both have the form $r\,\Delta(d/dr) + \frac{1}{2}\Delta'(d/dr)$ for an even quartic $\Delta$, the Laplace image of a maximal cut with exponent $-1/2$. Here $\Delta$ is four times the Gram determinant of the loop momenta of the tetrahedron, with every line on shell; that tetrahedron is the Laplace transform of the master integral [47]. The geminal's exponent is the parameter: as it goes to zero, $\rho \to 0$ and the operator falls to the Bessel one, as $N \to K_0$ above. Both densities of the master lie on one pair of centers, so for this operator the kernel does change the class: the same-pair collapse is a property of the Coulomb kernel. Symmetric exponent patterns, the James–Coolidge and Kołos–Wolniewicz ones and the single-exponent $(AA|AA)$ and $(AB|AB)$ classes, give $\rho = 1$, the equal-scale collapse of Eq. (5); the $(AA|BB)$ and $(AA|AB)$ patterns and unequal exponents give a genus-one operator. Caps: this classifies the master's operator, not the physical integrals, which are its derivatives in the exponents; the master's source term was not re-derived, though the master's published values at its two test points were reproduced to $10^{-15}$ by an independent momentum-space route; no earlier statement of the identification was found in a short search (record `docs/predictions/2026-10-27_h2_geminal_elliptic.md`, registered before computing; guard `tests/test_paper59_h2_geminal_master.py`).

[**OBSERVATION + PRIOR ART**] Read as a Feynman graph, the cases of this section and of Sec. VIII D fit one pattern. The densities and the kernels are the lines; a Slater geminal is a line whose mass is its exponent $\gamma$, and the Coulomb kernel is a line with no mass. On every instance tested the class is set by the shape of the graph together with the masses on its lines. A massless Coulomb line adds no threshold, so for Coulomb integrals the density topology alone sets the rung, as stated above. A Slater geminal keeps the curve the densities build, but it lifts the same-pair master above to genus one, and the correlated triangle's operator drops order where the geminal's mass equals that of a density pair, $\gamma = 1$. A Gaussian geminal is not a propagator of this kind, and its integrals leave the Picard–Fuchs ladder (Sec. VIII D). That the internal masses set the geometry of a Feynman integral is standard: the sunset graph with massless lines evaluates in Gamma functions, and with massive lines it is elliptic [28, 57]. What is recorded here is only that these molecular integrals follow the same dictionary. The nearest theorem in shape is Brown's: for massless primitive $\phi^4$ graphs in four dimensions the set of periods is minor monotone, so a minor-closed class of graphs bounds the ring the periods lie in [27]. That theorem has no masses; here the masses enter too. It is a pattern across the tested instances, not a derivation. It would fail on a Coulomb-only integral whose class depends on anything but the density topology, or on a Slater-geminal integral whose class does not change at an equal-mass value of its exponent (guards `tests/test_paper59_h2_geminal_master.py`, `tests/test_paper59_geminal_shapes.py`, `tests/test_paper59_gy_gaussian.py`,

tests/test_paper59_mg_mechanism.py).

## VIII. BEYOND THE THIRD CENTER: A FOURTH CENTER, A THREE-ELECTRON CHAIN, FOUR LINKED DENSITIES, AND A RING

Two results pre-registered on 2026-10-13 (records docs/predictions/2026-10-13_four_centre_genus.md and docs/predictions/2026-10-13_three_electron_k3.md, each committed before its computation) extend the genus reading of Sec. III in two directions: more centers, and more Coulomb kernels. A third, registered on 2026-10-16 (docs/predictions/2026-10-16_linking_ceiling.md, written before its computation but not committed beforehand), asks how the rung grows with the number of linked densities. A fourth, registered and committed on 2026-10-17 before its computation (docs/predictions/2026-10-17_density_loop.md), asks whether a ring of linked densities climbs faster than a tree. A fifth, committed the same day before its computation (docs/predictions/2026-10-17_ring_extensions.md), asks whether the correlated triangle, with geminals on two edges, behaves like the Coulomb one, and how a ring of four behaves; the second test stopped for feasibility. A sixth, committed before its computation (docs/predictions/2026-10-17_ising_class.md), asks where the Ising-class moments sit on the ladder.

### A. A fourth distinct center adds no genus

[**MEASURED**] The general four-center integral $(XY|ZW)$, four distinct centers and $1s$ Slater orbitals, reduces by the same Feynman parametrization to one radial $k$-integral over two parameters $(s,t)$ whose integrand carries exactly the two dispersion roots $\Delta_1(s)$, $\Delta_2(t)$ of Eq. (5); the centers enter only through a single phase vector $\boldsymbol{W}(s,t)$ and the two pair distances $D_1$, $D_2$. The reduced formula gives 0.218814149639 against 0.218814149425 from McMurchie–Davidson on sixteen-Gaussian fits (the residual is the fit gap), and reproduces the three-center value 0.204941652531 of Sec. II as a special case. There is no genus two at four centers, consistent with the class being set by the number of distinct two-center densities the kernel couples, not by the number of nuclei; this is that rule's first instance beyond the one this paper was written on. Its honest weight is that of a corollary of Sec. II, a calibration rather than a hard test. The momentum form itself is the Averys' (Ref. [4], App. B); the genus reading is not in it. Guards: tests/test_paper59_four_centre_genus.py (both values recomputed against McMurchie–Davidson; the kernel derived symbolically as the second $z$-derivative of a Yukawa product; the integrand's square roots and the dependence on the pair placement checked).

### B. A three-electron chain lives on a K3 surface

[**SYMBOLIC + MEASURED**] The chain integral $\langle\rho_a(1)\rho_b(2)\rho_c(3)/(r_{12}r_{13})\rangle$, three two-center $1s$ Slater transition densities (dispersion roots $\Delta_b=\sqrt{c_bp^2+1}$, $\Delta_c=\sqrt{c_cq^2+1}$, $\Delta_a=\sqrt{c_a|p+q|^2+1}$, scales $a=c_a^{-1/2}$, $b=c_b^{-1/2}$, $c=c_c^{-1/2}$) joined by two Coulomb kernels at electron 1, has two momentum integrations and three dispersion roots. The angular integral is elementary (no logarithm); the reduced two-dimensional fibre matches the direct three-dimensional one ($10^{-15}$ to $4\times10^{-14}$ in the record, $10^{-9}$ in the reduced-precision guard); the one-center limit is rational (23/54, 133/144, 289/225 at the three registered exponent triples); the branch locus in $\mathbb{P}^2$ is six lines with no four concurrent, so the double cover is a K3 surface, and fixed-$p$ slices are non-isotrivial genus-one curves. [**MEASURED / classical**] The holomorphic period reduces by one Fourier transform to the three-scale Bessel moment $\bar V_3=\int_0^\infty K_0(at)K_0(bt)K_0(ct)\,dt$ (the notation of Ref. [21]), whose closed form for $c\ge b\ge a>0$ is $(2c/\pi)\,\bar V_3=K(k_-)K(k'_+)+K(k_+)K(k'_-)$, with the moduli $k_\pm$ of Ref. [21] (Eq. 33, obtained there from an identity of Bailey [22]), checked at four scale triples to $10^{-34}$ in the record and at three in the guard to $10^{-17}$. The K3's period is therefore "elliptic squared": it opens no new transcendence class, being a sum of products of periods of the class of Sec. III. The reduction from a Slater chain to $\bar V_3$ is this paper's; the closed form is that of Ref. [21]. In the language of Sec. VI, the repulsion integral sits on the rung of the two-loop sunrise and the chain on the rung of the three-loop banana; the families differ, and Sec. VIII C gives the exact relation. [**OBSERVATION, evidence**] At equal scales the true fibre at zero separation collapses to two K3 periods (Bessel moments of the record) and $\pi^3$. At four generic scale triples, $(1;2,4)$, $(1;3,7)$, $(3;1,7)$ and $(3;4,6)$ with the central density's scale first, integer-relation searches at 340 and 400 digits find no relation in five bases built from the K3 periods and their scale derivatives, products of elliptic periods at the Bailey moduli, residue periods and their products, dilogarithms and trilogarithms of scale ratios, the vacuum sunrise and its mass derivatives, logarithms times periods, and weight-three Tate numbers. Every basis recovers a planted relation, and in every basis the fibre with $\int_0^x K_0$ dropped from each leaf reduces onto the K3 periods; three of the twenty searches are void, their precision cap below the planted relation's height. Moving the central density within the scale set $\{1,3,7\}$ gives no relation either, and the fibre does not collapse at the two threshold triples $(2;1,1)$ and $(3;1,2)$, where the central scale is the sum of the others. That is evidence, at a sample of triples and with guessed bases, that the generic fibre carries content beyond the K3 and elliptic periods, as the banana does; on a one-parameter slice it is a K3 function integrated once (Sec. VIII D), a class no finite basis of periods is

expected to capture. **[SYMBOLIC + MEASURED]** A second slice, with scales $(t;1,2)$, passes through two of these triples, $(1;2,4)$ at $t=1/2$ and the threshold triple $(3;1,2)$ at $t=3$, where the fibre values agree with the slice to 20 digits. Its Mellin operator has order 17 and divides exactly over the factor $\theta+2$ and two first-order pieces, one for each leaf, which enter together as their least common left multiple. Above them sits a piece of order four, equivalent by an exact map with poles only at $t=0,\pm1,\pm3$ to the operator of the three-scale moment $\int_0^\infty K_0(tx)K_0(x)K_0(2x)\,dx$. The composite of order seven annihilates the fibre to $5\times10^{-33}$ of its largest term at 32 digits, the fibre needs the order-four piece, and a search modulo a prime finds no rational map splitting it off within the bound set by the local exponents, while the same call finds a planted split. So on this slice too the fibre is an elementary function plus a K3 function integrated once, which accounts for the null at those two triples (record `docs/predictions/2026-10-21_generic_slice.md`). Longer trees are taken up in Sec. VIII C. Guards: `tests/test_paper59_three_electron_k3.py` (the fibre agreement, the elementary angle integral, the rational one-center limits, the closed form against quadrature, the K3 period against $\bar V_3$, the six-line branch locus, the non-isotrivial slice); `tests/test_paper59_generic_slice.py` (the second slice); the equal-scale collapse and the generic null are not guarded, and the records carry them (`docs/predictions/2026-10-13_three_electron_k3.md`, `docs/predictions/2026-10-19_p3e2_widened.md`).

### C. Four linked densities: the first rung beyond elliptic

**[SYMBOLIC + MEASURED]** The holomorphic period of a tree of linked densities depends only on how many densities the kernels link, not on the pattern of the links; a Coulomb ring of three behaves differently (Sec. VIII D). Take $n$ two-center $1s$ densities joined into a tree by $n-1$ rational kernels, and write the dispersion root of density $v$ at the fingerprint fibre as $\Delta_v=\sqrt{c_vL_v^2+1}$, with $L_v$ the signed sum of the momenta on the kernels at $v$. Fourier-transforming each $1/\Delta_v$ to $K_0(|y|/\sqrt{c_v})/(\pi\sqrt{c_v})$ turns every momentum integration into a delta function that identifies the two ends of its kernel, and a tree is connected, so

$$\int_{\mathbb{R}^{n-1}} d^{n-1}k\,\prod_{v=1}^{n}\frac{1}{\Delta_v(L_v(k))}=\frac{2^n}{\pi\sqrt{c_1\cdots c_n}}\int_0^\infty\prod_{v=1}^{n}K_0(a_vx)\,dx,\qquad a_v=c_v^{-1/2},\tag{15}$$

for a star and a chain alike. At $n=3$ this is the $\bar V_3$ reduction of Sec. VIII B; at $n=2$ it is the elliptic period of Eq. (6). A three-dimensional evaluation of four-density stars and chains, with the faster-decaying roots $\Delta_v^{-3}$ to which the same argument applies, matches the Bessel-moment side to between $4\times10^{-9}$ and $2\times10^{-7}$, the grid's own convergence. That check tests the identity's bookkeeping; the real three-dimensional integral reduces to moments of the same family with other weights in $x$, as shown below for both trees of four densities, the star and the path.

**[SYMBOLIC]** To read the rung, vary one scale: $F_n(t)=\int_0^\infty K_0(x)^{n-1}K_0(tx)\,dx$. The Mellin transform of $K_0^{n-1}$ obeys the recurrence of the $(n-1)$-th symmetric power of the Bessel operator, and that of $K_0(tx)$ is $2^{s-2}\Gamma(s/2)^2$, so $F_n$ satisfies an explicit linear equation in $t$ of order $n$; **[MEASURED]** each one is checked against direct quadrature of $F_n$ to $10^{-28}$. For $n=2$ it is the Legendre equation of $K$. For $n=3$ it is the equation of ${}_3F_2(\frac12,\frac12,\frac12;1,1;t^2/4)$, Clausen's square of an elliptic period, as Bailey's formula requires. For $n=4$, in the variable $x=1/(4t^2)$, it is

$$\theta^4-x(2\theta+1)^2(10\theta^2+10\theta+3)+9x^2(2\theta+1)^2(2\theta+3)^2,\qquad \theta=x\,\frac{d}{dx},\tag{16}$$

which has maximally unipotent monodromy at $x=0$ and conifold points at $x=1/36$ and $x=1/4$, with local exponents $(0,1,1,2)$. Its holomorphic solution is $\sum_m B_m(x/4)^m$ with integer $B_m=\binom{2m}{m}^2\sum_k\binom{m}{k}^2\binom{2k}{k}$, the term-by-term (Hadamard) product of $\binom{2m}{m}^2$ with the Apéry-like sequence of the equal-mass sunrise, and its mirror map has integer coefficients.

**[SYMBOLIC]** Whether a family is built from elliptic periods shows in the quadratic relations among the solutions of its equation: for monodromy of these types, an elliptic family has none, the square of one elliptic family or the tensor product of two has one, and the cube of one has three. Counted exactly on Taylor bases at two rational points, the numbers are $0,1,0,1$ for $n=2,3,4,5$. For $n=4$ the operator is irreducible, with no bound on degrees: a right factor of order $k$ would carry, at each singular point, $k$ of the operator's local exponents, and Fuchs' relation then fails for every $k<4$, however many apparent singularities the factor adds. The same count shows the operators for $n=2,3,5$ irreducible. Its Yukawa coupling, $1+16q+1152q^2+99520q^3+\cdots$, is not constant, as it would be for a symmetric cube. An irreducible fourth-order local system with no invariant quadric that is not a symmetric cube has differential Galois group $\mathrm{Sp}_4$ or $\mathrm{SL}_4$, and neither is a quotient of a product of copies of $\mathrm{SL}_2$. So $F_4$ is not a polynomial in elliptic periods with algebraic moduli, and neither is the generic four-scale moment, which restricts to it. Two linked densities give an elliptic period, three give a K3 whose period is elliptic squared, and four give the first period of a new kind. At $n=5$ the single relation marks an orthogonal structure of rank five, not the fourth power of an elliptic period, which would carry six. That is the structure of the exterior square of a fourth-order Calabi–Yau operator, and the tables of Ref. [52] print the operator of $F_5$ verbatim, attached to entry #39, a different operator from the #58 of $n=4$. So five linked densities give no new type of period: the type is that of four, in a different family.

[**OBSERVATION**] Every electron-repulsion integral couples exactly two densities, so the integrals of a molecular Hamiltonian stay on the elliptic rung however large the molecule. The class climbs only in correlated integrals that link more densities, such as the four-electron bridge integrals of explicitly correlated methods taken over two-center densities. Two caps apply. These are statements about the fibre and its holomorphic period, so the integrated value may sit lower, as the two-center exchange class does (Sec. VII). And the reduction of the real three-dimensional integral into the family of Eq. (15) is shown only for the chain of three and the two trees of four below; for five or more densities it is not shown. [**SYMBOLIC + MEASURED**] At the fingerprint fibre on the slice $(t;1,1,1)$ the star is $(16/\pi^4)\,t^3\int_0^\infty \chi(x)^3 K_0(tx)\,dx/x$, with $\chi=\int_0^x sK_1(s)\,ds$ (Sec. VIII D), checked against a momentum-space evaluation to $2\times10^{-21}$. Its Mellin operator, of order 19, divides exactly with a piece of order four equivalent to the operator (16) by an exact rational map. An order-eight right factor annihilates the star to $3\times10^{-45}$ of its largest term at 45 digits; its composition factors are that piece, the Legendre operator of $F_2$ and two of order one, with no K3 piece, and the star needs the top piece. So the star sits on the rung of four linked densities and skips the K3 rung; minimality is bounded by a search modulo a prime, not proved (record `docs/predictions/2026-10-21_four_density_star.md`). Guards: `tests/test_paper59_four_density_rung.py`, `tests/test_paper59_four_density_star.py`. [**SYMBOLIC + MEASURED**] The path, four densities in a chain, reduces the same way. On the slice $(1,t,1,1)$, with $t$ on an inner density, it is $(16/\pi^4)\,t^3\int_0^\infty\int_0^\infty \min(x,y)\,K_0(tx)\chi(x)\,K_0(y)\chi(y)\,dx\,dy$, checked against a momentum-space evaluation to 20 digits. Its holonomic relations give an operator of order 74 with an exact right factor of order 12, which annihilates the path to below $10^{-28}$ of its largest term at 32 digits and $4\times10^{-56}$ at 60. That factor splits exactly into a piece equivalent to the operator (16), three pieces equivalent to the Legendre operator of $F_2$, and two of order one, each equivalence by an exact rational map; there is no K3 piece, and the path needs the top piece. So both trees of four densities sit on the same rung and skip the K3 rung; for the path, minimality is bounded by a search modulo a prime up to $t$-degree 260 (record `docs/predictions/2026-10-22_four_density_path.md`). Guard: `tests/test_paper59_four_density_path.py`.

[**PRIOR ART**] Most of this structure is known in the mathematics of Bessel moments; what is new here is the route into it from a molecular integral. The operator (16) is entry #58 of the Calabi–Yau tables of Almkvist, van Enckevort, van Straten and Zudilin [52] (entry 2.9 of the Calabi–Yau operator database [53]), listed there as a Hadamard product, with the same holomorphic coefficients; Gegelia and van Straten [54] list it among their conjectural sources of paramodular forms, at conductors 2145 and 2465. For every $n$ from 2 to 5 the holomorphic solution of the equation of $F_n$ at $t=\infty$ is $t^{-1}\sum_m \binom{2m}{m}^2 W_{n-1}(2m)\,(16t^2)^{-m}$, the Hadamard product of $\binom{2m}{m}^2$ with the even moments $W_{n-1}(2m)$ of the $(n-1)$-step planar random walk. The Hadamard construction, and the tables that place its $n=4$ and $n=5$ cases at entries #58 and #39, are Almkvist and Zudilin's [56]; the links of $W_N$ to Bessel moments, random walks and Calabi–Yau equations are Broadhurst's [64]; Vanhove's operators for the equal-mass two-dimensional banana integral with $N$ propagators [57] annihilate $\sum_k W_N(2k)s^{-k-1}$, checked here in exact arithmetic for $N=2$ to 5, so that series is the banana's holomorphic period at large $s$. The alternation of the monodromy, symplectic for $n$ even and orthogonal for $n$ odd, matches the self-duality of the symmetric-power Kloosterman motives of Fresán, Sabbah and Yu [37], whose pairing is symplectic when the number of Bessel factors is even and orthogonal when it is odd. [**SYMBOLIC**] So the tree ladder is tied to the banana ladder, where an $l$-loop equal-mass banana integral carries a Calabi–Yau $(l-1)$-fold [67], by one exact operation: with $y=1/(16t^2)$ and $F_n=t^{-1}g(y)$, the operator of $g$ is the Hadamard product of the Legendre operator with the operator of the banana with $n-1$ propagators, checked in exact arithmetic for $n=2$ to 5. Since $\int_0^\infty J_0(px)\,dp=1/x$, the same fact reads $F_n(t)=\int_0^\infty dp\int_0^\infty x\,J_0(px)\,K_0(tx)K_0(x)^{n-1}dx$: $F_n$ is the $n$-propagator banana, with masses $(t,1,\ldots,1)$, integrated over a line of external momentum. It sits on the rung of the banana with $n+1$ propagators but is not that family: its singular values of $t^2$, $\{1\},\{4\},\{1,9\},\{4,16\}$ for $n=2,\ldots,5$, are the thresholds of the banana with $n-1$ propagators.

[**SYMBOLIC + MEASURED**] At $t=1$ the ladder is the Ising class of Bessel moments, $F_n(1)=c_{n,0}=\int_0^\infty K_0(x)^n\,dx$ [21]. The singular values of $t^2$ are the thresholds $(n-1-2j)^2$ for every $n$ up to 8, with no other finite nonzero singular point, so $t=1$ is a singular point of $F_n$'s operator exactly when $n$ is even; the operators for $n=6$, 7 and 8 are irreducible by the exponent argument above, so these are the singular points of $F_n$'s minimal operator. There one local exponent is doubled, while at the thresholds of odd $n$ one is a half-integer. Since $F_n$ is analytic for $t>0$, at a threshold it lies among the solutions without the logarithm or the branch. A prediction registered and committed before the computation (`docs/predictions/2026-10-17_ising_class.md`) read this as a rule for values: odd moments stay on their rung and even ones drop. The registered value cases hold, with moments for $n=2$ to 8 to 286 digits by two routes and every informative integer-relation search controlled; the registered attachment of $c_{4,0}$ to the conifold fibre was not tested. $c_{3,0}$ is a complex-multiplication elliptic value; $c_{5,0}$ and $c_{7,0}$ fit no tested lower basis, at coefficient bounds of $10^{14}$ to $10^{17}$ and $10^7$ to $10^9$; $c_{4,0}=4\pi^2L(f,2)$ with $f=\eta(2\tau)^4\eta(4\tau)^4$ of weight four and level eight, checked to $5\times10^{-261}$; and $c_{6,0}=108\pi^2L(g,4)$ with $g$

of weight six and level 24, to the same precision. But $c_{8,0}$ fits no single $L$-value tried, with no Tate relation to coefficient bound $10^7$ and none with an $L$-value to $10^{12}$, so the $n = 8$ search gives parity no support as a rule for values. [**PRIOR ART**] The $L$-value forms of $c_{3,0}$ and $c_{4,0}$ are collected by Lim, Tu and Yu [60], who credit them to earlier work, through the hypergeometric forms of Ref. [21]; the numerical form of $c_{6,0}$ is theirs, and in their account of the underlying Kloosterman motives the moments at $n = 5$, 7 and 8 enter only through $2 \times 2$ determinants of periods. The singular points themselves are the thresholds and pseudo-thresholds of the equal-mass banana operator (Ref. [67], Eq. (3.7)); the threshold reading of the values is what the ladder adds. In the bookkeeping of Papers 18 and 34 [70, 71] these values lie beyond the genus-graded embedding tier: $c_{3,0}$ is a rational multiple of $2^{-2/3}\Gamma(1/3)^6/\pi$, a complex-multiplication period, and $c_{4,0}$ and, numerically, $c_{6,0}$ are $\pi^2$ times critical $L$-values of modular forms, periods of the Kloosterman motives. The nearest physics analogue of a ceiling set by the calculation rather than the system is the bound Bourjaily, McLeod, von Hippel and Wilhelm argue for on the rigidity of massless four-dimensional Feynman integrals, $2(L-1)$ at $L$ loops [65]: there the loop order caps the geometry, here the number of densities one integral links. On the chemistry side, one-center three- and four-electron correlated integrals have closed forms [8, 66], consistent with the rung needing two-center densities. A targeted search found no source placing a molecular integral on these operators; that is an absence from a search, not a proof of absence. Matuzas [55] relates one channel of a different four-electron correlation integral, a ring of Coulomb links with one diagonal over Slater functions, to another Calabi–Yau operator, AESZ-330, by an inhomogeneous equation.

### D. A ring of three densities: the loop does not climb

Every rung above is a tree. An integral sits on a rung when the minimal annihilator of its fibre has no composition factor above that rung. A ring of kernels leaves one momentum that the delta functions behind Eq. (15) do not remove, and one might expect a loop to raise the class faster than a leg does. The test case is the Coulomb triangle $T = \langle \rho_a(1)\rho_b(2)\rho_c(3)/(r_{12}r_{13}r_{23})\rangle$, three densities with every pair joined by a kernel. Its explicitly correlated relative $\langle f_{ik}f_{jk}/r_{ij}\rangle$ is the one three-body class of the LiH correlated engine that does not separate. The prediction, written and committed on 2026-10-17 before the computation (`docs/predictions/2026-10-17_density_loop.md`), was that the loop does not climb.

[**SYMBOLIC + MEASURED**] The loop integrates out exactly. In three dimensions three massless propagators of unit index are unique, and the uniqueness (star-triangle) relation gives

$$\int \frac{d^3l}{|l-x_1|^2\,|l-x_2|^2\,|l-x_3|^2} = \frac{\pi^3}{|x_1-x_2|\,|x_2-x_3|\,|x_3-x_1|}, \tag{17}$$

which a Feynman-parameter evaluation confirms at five random triples to $1.2 \times 10^{-12}$. The triangle becomes an integral over the same two momenta and three densities as the three-electron chain. For spherical densities the magnitude measure cancels the weight, and

$$T = \frac{1}{\pi}\int_{|p-q|\le w\le p+q} \tilde\rho_a(p)\,\tilde\rho_b(q)\,\tilde\rho_c(w)\,dp\,dq\,dw. \tag{18}$$

[**SYMBOLIC + MEASURED**] At one center, with $1s$ Slater densities, Eq. (18) reproduces Remiddi's closed form [9], as written in Ref. [20] and differentiated in the exponents, to $2 \times 10^{-21}$. Where the chain is rational (Sec. VIII B), the triangle carries a dilogarithm. At exponents $(1,1,1)$

$$T = \frac{11}{3} - \frac{\pi^2}{8} - \frac{3}{2}\,\mathrm{Li}_2\!\left(-\tfrac{1}{2}\right) - \frac{46}{9}\ln 2 - \frac{3}{4}\ln^2 2 + \frac{3}{2}\ln 2\ln 3, \tag{19}$$

and at $(1,2,3)$ a 100-digit integer-relation search finds a closed form in $\mathrm{Li}_2(1/3)$, $\mathrm{Li}_2(1/5)$, $\mathrm{Li}_2(1/6)$, $\pi^2$ and products of logarithms. The ring raises the weight from zero to two. In the bookkeeping of Papers 18 and 34 these one-center values are embedding-tier constants of weight two, one step above the weight-one two-center seed set; the $\pi^3$ of Eq. (17) is the loop's normalization.

[**SYMBOLIC + MEASURED**] At the fingerprint fibre, $\tilde\rho_v = \Delta_v^{-3}$ with scales $(1,1,t)$, the triangle is a moment of three Struve functions,

$$G(t) = \frac{\pi}{2}\,t^2\int_0^\infty x^2\,\psi(x)^2\,\psi(tx)\,dx, \qquad \psi = \mathbf{L}_{-1} - I_1, \tag{20}$$

where $\psi$ is one density with its kernel in position space and solves $(\theta^2 - 1 - x^2)\psi = -2/\pi$. The Mellin route of Sec. VIII C gives

$$L_G = -4\theta(\theta-3)^2 + t^2\,(5\theta^3 - 7\theta^2 - 5\theta + 3) - t^4\,\theta(\theta+2)(\theta+3), \qquad L_G\,G = \tag{21}$$

with $\theta = t\,d/dt$, checked by quadrature of the Struve form at six values of $t$ to 18 digits, and independently on the momentum form, with the $t$-derivatives taken analytically. The source term is the operator applied to a constant: $L_G\,1 = 3t^2$, so $L_G(G+1) = 0$, and the constant is what the regularized Mellin transform cannot see. This operator factors. The solution of $L_G$ regular at $t = 0$ is algebraic, $(1 - t^2/4)^{-3/2}$. Write $L_G = M\,(\theta - \lambda_G)$ with $\lambda_G = 3t^2/(4 - t^2)$; the division is exact. The quotient $M$ is equivalent to the Legendre equation $\theta^2 - t^2(\theta+1)^2$ of $F_2$, the two-density period: the map $f \mapsto t^3(1+\theta)f/(4-t^2)$ sends every solution of the Legendre equation to a solution of $M$, an identity checked in exact rational arithmetic. And $G$ needs the elliptic piece: $(\theta - \lambda_G)(G+1)$ is a nonzero solution of $M$, equal to $-0.161$ at $t = 0.5$ and

$-2.33$ at $t = 1.3$. Neither side has a relation to the K3 family of $F_3$. So $G$ is built from an elementary constant, an algebraic function and an elliptic period integrated once against an algebraic kernel. The triangle sits on the elliptic rung of a single repulsion integral. That is one rung below the three-density chain.

**[SYMBOLIC + MEASURED]** The chain $C(t)$ at the same fibre, treated the same way, is the control. The Mellin route gives it an operator of order seven, which divides exactly as $N_2\,R_3\,R_1\,(\theta-2)$. The two lowest pieces are elementary: $\theta-2$ has the solution $t^2$, and $R_1 = (\theta-3)-t^2(\theta+1)$ has the solution $t^3/(1-t^2)^2$. The piece $R_3$, of order three, is equivalent to the operator $4\theta^3-t^2(\theta+1)^3$ of $F_3$: the map

$$f \;\mapsto\; \frac{t^3(t^4-18t^2-4)\,f + t^3(2t^4-43t^2-40)\,\theta f + t(t^6-27t^4-96t^2+32)\,\theta^2 f}{(t^2-4)^2} \tag{22}$$

sends every solution of $F_3$'s operator to a solution of $R_3$, an identity checked in exact rational arithmetic. The top piece $N_2$, of order two, is not needed: $R_3R_1(\theta-2)$ already annihilates the chain, with a residual of at most $2\times10^{-31}$ at 32 digits and $4\times10^{-19}$ at 20 digits at four values of $t$ between 0.5 and 3.5, while changing one coefficient of $R_3$ gives residuals from 1.9 to 604. The chain's only piece beyond the elementary ones is therefore the K3 family of $F_3$. Finer still, $R_3R_1$ splits exactly, so the order-five operator is a left multiple of an operator $P$ of order four, and the chain satisfies $P$: $PC$, divided by the solution $t^3/(1-t^2)^2$ of $R_1$, is at most $10^{-23}$ at 20 digits and $10^{-33}$ at 30 digits at four values of $t$, and $9\times10^{-45}$ at 40 digits at one, while a 1% change in one coefficient gives 0.02 to 0.07; the next paragraph derives this. The second solution of $R_1(\theta-2)$, which carries $\operatorname{artanh} t$, is absent. So $C = c\,t^2 + t^2\int^t s^{-3}(B'f)(s)\,ds$ with $f$ a solution of $F_3$'s operator and $B'$ a rational map. That integral cannot be done inside the K3 family: no rational map splits the K3 piece off $t^2$. This is proved by bounding the poles and degree of any such map from the local exponents and searching that bound exhaustively in exact arithmetic, a search that finds every planted split. Because $F_3$'s operator is irreducible, the chain is therefore not a rational combination of $F_3$ and its derivatives plus any function without a K3 piece: it is a K3 function integrated once against an elementary kernel. So the Coulomb ring lowers the rung at the fibre and raises the weight at one center. In neither limit does it climb the period ladder.

**[SYMBOLIC + MEASURED]** The same holds for the physical weights. With the Slater densities of Sec. VIII B, the chain's fibre at zero separation, on the slice with scales $(t;1,1)$ and $t$ on the central density, is

$$G(t) = \int_0^\infty x\,K_1(tx)\,J(x)^2\,dx, \qquad J = 3\int_0^x K_0 - 3xK_0 - x^2K_1 \tag{23}$$

The Mellin route gives an operator of order seven, which divides exactly as $N_2\,R_3\,R_1\,(\theta+2)$. The two lowest pieces are elementary: $\theta+2$ has the solution $t^{-2}$, and $R_1 = (1-t^2)\theta-(1+7t^2)$ has the solution $t/(1-t^2)^4$. The piece $R_3$, of order three, is equivalent to $F_3$'s operator $4\theta^3-t^2(\theta+1)^3$: an explicit order-two map, rational in $t$ with poles at $t = 0, \pm2$, sends every solution of $F_3$'s operator to a solution of $N_2R_3$, and $R_3$ is the annihilator of the image, both checked in exact rational arithmetic. Again $N_2$ is not needed: $R_3R_1(\theta+2)$ annihilates $G$ with a residual of at most $2\times10^{-31}$ at 32 digits at five values of $t$ between 0.5 and 3.5, while changing one coefficient of $R_3$ by one part in 198 gives residuals from $10^3$ to $4\times10^4$. And $G$ needs the K3 piece: $R_1(\theta+2)G$ is nonzero, 191 at $t = 0.5$ and 8.9 at $t = 3.5$. A search modulo a prime finds no rational map that splits the K3 piece off, in a box larger than the bound set by the leading terms of the local exponents, and the same call finds a planted split. So $G$ is an elementary function plus a K3 function integrated once against an elementary kernel. The second solution of $R_1(\theta+2)$, which carries $\operatorname{artanh} t$, is absent, as it is for the chain. In $w = t^2$, $\delta = w\,d/dw$, the exact system $R_1S = 1$ modulo $R_3$ has a unique rational solution $S = s_0 + s_1\delta + s_2\delta^2$ inside the bound set by the local exponents, with denominator $w(w-1)^4(2w+7)$. So $R_3R_1(\theta+2) = Q_1P$ with $P = (\theta+2) - S\,R_1(\theta+2)$ of order four, and $P$ kills $t^{-2}$. On $G$, $P$ leaves at most $2\times10^{-26}$ of its largest term at 25 digits at $t = 1/2$ and $13/10$, while a 1% change in $s_0$ gives $10^{-3}$. Both annihilations are derived, not only measured. With $W = R_3R_1(\theta+2)$, the Mellin operator is $N_2W$, and $N_2WG = 0$ holds with no source term. Each term $t^{2i}Q_i(\theta)G$ of that operator is an integral on the line $\operatorname{Re} s = c-2i$, $2 < c < 4$; moving it to $c$ crosses the poles of the Mellin transform of $G$ at $s = 4, 6, 8$, which come from the series of $J^2$ at $x = 0$. The six residues cancel exactly, though none of them vanishes alone. The recurrence itself has no boundary term, since the relation for $J^2$ kills constants. The same bookkeeping gives the nonzero sources $-16/t$ for $\int_0^\infty xK_1(tx)J(x)\,dx$ and $-9216\,t^2$ for $\int_0^\infty xK_0(tx)J(x)^2\,dx$, both equal to quadrature at 25 digits. So $y = WG$ solves $N_2y = 0$. In the normal form $v'' + qv = 0$ of $N_2$, $q = 6 - 59/(4t^2) + O(t^{-4})$, with $q \geq 5$ for $t \geq 3.42$ and $q' = O(t^{-3})$. So by Sturm comparison every nonzero real solution has infinitely many zeros there, and its amplitude does not fall faster than $t^{-1/2}$. A function with an expansion in powers and logarithms of $t$ at infinity has neither property, and $y$ has one; so $WG = 0$. For the chain the residues at $s = 0, 2, 4$ cancel the same way, and $q \to 20/9$ with $q \geq 2$ for $t \geq 5.32$, so $WC = 0$ for its $W = R_3R_1(\theta-2)$. Since $R_1S = 1 + NR_3$ with $N$ rational, $R_1P = -N\,R_3R_1(\theta+2)$, so $P$ sends every solution of $R_3R_1(\theta+2)$ to a multiple of $t/(1-t^2)^4$, which is odd in $t$. Near $x = 0$, $J$ is a sum of odd powers of $x$ times powers of $\log x$. So to the right of its strip the Mellin transform of $G$ has poles only at $s = 4, 6, 8, \ldots$, and the expansion of $G$ at large $t$ has only even powers of $t$, times logarithms. The coefficients of $P$ are rational in $t^2$, so $PG$ has the same property. On $t > 1$, where $R_1$ is regular, $PG$ is a single multiple of $t/(1-t^2)^4$, so that multiple is zero. Since $G$ is analytic on $\operatorname{Re} t > 0$ and $P$

has rational coefficients, $PG$ is meromorphic there; it vanishes on $t>1$, so $PG=0$ for all $t>0$. Equivalently, among the solutions of $R_3R_1(\theta+2)$, $\ker P$ is the set that are even in $t$ at infinity, and at infinity $P$ keeps exactly the integer exponents of $R_3R_1(\theta+2)$, in $w$. The analyticity of $G$ at $t=1$ alone, or at $t=2$ alone, does not give $PG=0$: $P$ has a solution with a logarithm at $t=1$ and one with a half-integer exponent at $t=2$. For the chain, $C=(8t^3/\pi^3)\int_0^\infty \chi(x)^2K_0(tx)\,dx$, where $\chi=\int_0^x sK_1$ is, near $x=0$, a sum of odd powers of $x$ times powers of $\log x$. So $\int_0^\infty \chi^2K_0(tx)\,dx$ has only odd powers of $t$ at infinity, times logarithms, and with the factor $t^3$ the function $C$ has only even ones; the same steps then run with the solution $t^3/(1-t^2)^2$ of its $R_1$. The operator $R_3R_1(\theta+2)$ is singular at $t=1$, the equal-scale point where the fibre collapses (Sec. VIII B); its exponent $-3$ there comes from $R_1$, so the singularity is not apparent. Caps: the no-split search is modulo a prime, and its bound rests on the leading-term argument rather than the written proof the chain has; the annihilation and the parity step use the standard Mellin contour shift. Records: `docs/predictions/2026-10-19_p3e2_widened.md`, `docs/predictions/2026-10-20_slice_artanh.md`, `docs/predictions/2026-10-20_slice_parity_proof.md` and `docs/predictions/2026-10-21_wg_closure.md`. Guard: `tests/test_paper59_chain_slater_slice.py`.

**[SYMBOLIC + MEASURED]** At $\gamma=1$ the correlated triangle does not drop. Put the exponential geminal $f=e^{-\gamma r}$ on the two edges at one density, $Y=\langle\rho_i\rho_j\rho_k\, f_{ik}f_{jk}/r_{ij}\rangle$, at the fibre with scales $(1,1,t)$ and $t$ on the density that carries both geminals. The registered prediction was that $Y$ stays on the elliptic rung; it fails. The loop is still elementary: it is a mass derivative of the loop with masses $(\gamma,\gamma,0)$, a solid angle, $\pi^2\arctan(\sqrt{\det\mathcal{G}}/\mathcal{D})/\sqrt{\det\mathcal{G}}$ in the Gram data $\mathcal{G}$, $\mathcal{D}$ of the masses and momenta, matched by direct integration at five triples to $3\times10^{-14}$. But it has no uniqueness step. The two densities and the two geminals around it form a kite, whose Taylor series satisfies an order-two operator, measured on 600 coefficients and mapped exactly to the Legendre equation: the kite is elliptic. At $\gamma=1$ the minimal annihilator of $Y$ has order four and divides exactly as $\tilde R_3\tilde R_1$, with factors of orders three and one that are not those of the chain. The piece $\tilde R_1$ has an algebraic solution. The piece $\tilde R_3$ is the adjoint of an operator whose solutions are the image of the solutions of $F_3$'s operator under an exact order-two map, and since $F_3$'s operator is anti-self-adjoint, $\tilde R_3$ is equivalent to it. Applied to $Y$, the order-four operator leaves at most $2.3\times10^{-26}$ of its largest term at four values of $t$, with no source term, and no operator of order three annihilates $Y$: an exact search within a bound set by the local exponents finds no split. Attaching the last density turns the elliptic kite into the K3 family, a Clausen-type step. So at $\gamma=1$ the correlated triangle sits with the three-density chain, on the K3 rung, and not with the Coulomb triangle. At $\gamma=3/2$ an annihilator of $Y$ of order nine has an exact composition series whose pieces have orders one (four of them, each with an algebraic solution), two and three. The order-three piece has the local data of $F_3$'s operator twisted by $t$, with genuine singular points only at $t=0,\pm2,\infty$, which fixes it up to equivalence by Levelt's rigidity, in the form of Beukers and Heckman [58]; an exact order-two map from $F_3$'s operator confirms the equivalence. The order-two piece is elliptic. In $x=t^2$ it is equivalent, by an exact map of order one, to the operator of a period of the rational elliptic surface $Y^2=X^3-3x^2(16x^2+152x+25)\,X-2x^3(4x-5)(16x^2-616x+25)$, whose $j$-invariant is $(16x^2+152x+25)^3/\big(x(4x-1)^2(4x-25)^2\big)$ and whose singular fibres are $I_1^*$, $I_2$, $I_2$, $I_1$ at $x=0,1/4,25/4,\infty$; over $t$ they are six fibres $I_2$, at $t=0,\pm1/2,\pm5/2,\infty$, that is at $t=|1-\gamma|$ and $1+\gamma$. The monodromy of the piece has integer traces, measured at 80 digits, and the map is exact (record `docs/predictions/2026-10-22_c2b_identification.md`).

**[SYMBOLIC + MEASURED]** The curve is Legendre's, with modulus $\lambda=(\gamma^2-(t-1)^2)/(4t)$, so that $\lambda(1-\lambda)$ is $-1/(16t^2)$ times the Källén function of $t^2$, 1 and $\gamma^2$. Twisted by $t$, the Legendre operator of degree $-1/2$ in $z=(t^2+1-\gamma^2)/(2t)$ is the period operator above, coefficient for coefficient. **[PRIOR ART]** The function itself is $\pi\bar S_3(a,b,w)$ of Ref. [21] (their Eq. 30), given there as a convolution of two square-root propagators, $(\pi/4)\int_{-\infty}^{\infty}dk/\sqrt{(k^2+a^2)((k-c)^2+b^2)}$, and as a complete elliptic integral of modulus $\sqrt{(c^2+(a-b)^2)/(c^2+(a+b)^2)}$; the Legendre form, the imaginary momentum and the triangle are the readings added here. It annihilates $C(t)=\int_0^\infty K_0(tx)K_0(x)\cosh(\gamma x)\,dx$, which converges for $t>\gamma-1$, the pseudo-threshold, to $5\times10^{-41}$ at 40 digits, and $C=\pi^2P_{-1/2}(z)/(4\sqrt{t})$ to $2\times10^{-16}$ at 15 digits: the continuation to imaginary momentum of a classical cosine transform of $K_0K_0$. Since $\int dp_2\,e^{ip_2x_2}=2\pi\delta(x_2)$, $C$ is $\pi$ times the two-dimensional bubble with masses $t$ and 1, integrated over a line of external momentum moved off the origin by $i\gamma$; at $\gamma=0$ it is $F_2$. In the dictionary of Sec. VIII C the geminal therefore acts as an imaginary external momentum, and the singular points are where $\gamma=t+1$ or $\gamma=|t-1|$, the bubble's threshold and pseudo-threshold. The registered candidate with $e^{-\gamma x}$ in place of $\cosh(\gamma x)$ is not a solution, because its sinh part is not (record `docs/predictions/2026-10-22_c2b_feynman.md`).

This fixes the operator class of the piece, not the component of $Y$ along it. **[SYMBOLIC + MEASURED]** At $\gamma=2$ the reading was registered before $Y$ was computed there, and it held piece by piece. An annihilator of order nine, fitted from 800 series coefficients and checked on $Y$ to $5\times10^{-26}$, has the same composition series as at $\gamma=3/2$. Its order-two piece is equivalent by an exact map to the Legendre operator with $z=(t^2-3)/(2t)$, its order-three piece is the twisted K3 piece at $t=\pm2$ again, and its order-one pieces are algebraic, branched at $t=2$ and $t=2\gamma=4$ (record

docs/predictions/2026-10-22_c2b_feynman_g2.md). **[SYMBOLIC + MEASURED]** Why the order drops at $\gamma = 1$. In Feynman parameters the kite's density pair enters through $M(u) = \sqrt{1 + u(1-u)z}$ and its geminal pair through $N(v) = \sqrt{\gamma^2 + v(1-v)z}$. At $\gamma = 1$ the two are the same function: the two bubbles have equal masses, and the four pair thresholds $t = 2$, $2\gamma$, $1 + \gamma$, $|1 - \gamma|$ fall onto 2 and 0. Two readings were registered and separated before computing: equal masses, or merely colliding thresholds. At $\gamma = 3$ the pseudo-threshold $|1 - \gamma|$ falls on the density-pair threshold $t = 2$ with unequal masses. There the annihilator of $Y$ keeps order nine and the series $1, 1, 1, 1, 2, 3$, the elliptic piece is present, mapped exactly from the Legendre operator with $z = (t^2 - 8)/(2t)$, and the genuine singular points are exactly $t = 2, 4, 6$; at $t = 2$ the K3 piece's half-integer exponents and the elliptic piece's $I_2$ fibre sit together, and no factor is lost. At $\gamma = 1/2$, before the branch point crosses the contour, the order is nine as well, with the Legendre map at $z = (t^2 + 3/4)/(2t)$ and singular points at $t = 1/2, 1, 3/2, 2$. So colliding thresholds remove nothing, and the drop at $\gamma = 1$ is the equal-mass degeneration, the step that makes the equal-mass sunrise special in the Feynman-integral literature [28]. The kite's operator in $z$ has order six at $\gamma = 1/2$, $3/2$, 2 and 3, with genuine singular points exactly at minus the squared pair masses and at 0 and $\infty$, and order two at $\gamma = 1$. The annihilators are checked on $Y$ to $6 \times 10^{-27}$ at $\gamma = 1/2$ and $3 \times 10^{-24}$ at $\gamma = 3$, and the maps are exact (record docs/predictions/2026-10-23_gamma1_mechanism.md; guard tests/test_paper59_mg_mechanism.py). Cap: the mechanism is read from the integrand and tested by its consequences at two values of $\gamma$; no symbolic derivation of the drop from $N = M$ is given. **[SYMBOLIC + MEASURED]** The cusp-correct geminal $f = re^{-\gamma r}$ of the LiH engine, on both edges at $\gamma = 3/2$ and 2, brings no new type, as registered before computing from the fact that $re^{-\gamma r}$ is a $\gamma$-derivative of the same Yukawa triangle. Its annihilator of order nine, checked on the integral to $10^{-28}$, has the same composition series: four algebraic pieces, branched at $t = 2$ and 3; an order-two piece, a different operator from the exponential case's, equivalent by an exact map to the same Legendre family; and the twisted K3 piece (record docs/predictions/2026-10-22_geminal_shapes.md; guard tests/test_paper59_ga1_geminal.py). At $\gamma = 2$, registered as a check of the rule, the order is nine again with the same series and the same degrees, the genuine singular points are exactly $t = 1, 2, 3, 4$, the order-two piece is again a different operator from the exponential case's and maps exactly from the Legendre operator with $z = (t^2 - 3)/(2t)$, and the algebraic pieces carry the exponents $-11/2$ and $-13/2$ at $t = 2\gamma$ as at $\gamma = 3/2$; the operator is exact, from coefficients modulo twelve primes and rational reconstruction, and leaves $2 \times 10^{-24}$ on the integral at 32 digits (record docs/predictions/2026-10-23_geminal_shapes_g2.md; guard tests/test_paper59_ga3_geminal.py). **[SYMBOLIC + MEASURED]** One function carries both elliptic pieces met so far. Write $\mathcal{L}(a, b; c) = \int_0^\infty K_0(ax)K_0(bx)\cos(cx)\,dx = \pi^2 P_{-1/2}(z)/(4\sqrt{ab})$, with $z = (a^2 + b^2 + c^2)/(2ab)$. At $c = 0$ it is the radial core of the three-center integral at zero separation: with $a_i = c_i^{-1/2}$ it reproduces Eq. (6) to $4 \times 10^{-31}$ at three pairs of scales. At $c = i\gamma$ it is $C(t)$ above. Zero or real momentum gives $z \geq 1$. Imaginary momentum gives $z = \cos\theta$ between the pseudo-threshold and the threshold, $|t - 1| < \gamma < t + 1$, with $\theta$ the angle opposite the side $\gamma$ in the triangle with sides $t$, 1 and $\gamma$, and the modulus is $\lambda = \sin^2(\theta/2)$. The thresholds are the flat triangles, since the Källén function is $-16$ times the squared area. The slot is not a convention: with the geminal as a mass and a density scale as the momentum, the two other assignments give the families $j = 4(16x^2 + 472x + 25)^3/\big(9x(4x-1)^2(4x-25)^2\big)$ and $j = 4(16x^2 - 104x + 601)^3/\big(9(4x-1)^2(4x-25)^2\big)$, and neither is isogenous to the piece's, since their traces of Frobenius differ at most good primes at every rational $x$ tried, while the two solutions of the degree-six search agree at every one. The right angle, $\gamma^2 = t^2 + 1$, gives $\lambda = 1/2$ and $j = 1728$, the same complex-multiplication value $K(1/\sqrt{2})$ as the three-center fibre at $\rho = 1/2$. **[OBSERVATION]** On this family, then, composing unequal decay rates costs one elliptic period: at zero momentum its modulus is set by their ratio, and equal rates make it elementary, as Eq. (6) records; at imaginary momentum it is set by the angle of their triangle, and a flat triangle is singular. Reading one-loop integrals through triangles of masses and momenta is the approach of Davydychev and Delbourgo [61]; what is added is its appearance in a correlated molecular integral. **[SYMBOLIC + MEASURED]** The shape of the geminal matters at this level. A Gaussian geminal's avatar, $\int_0^\infty K_0(tx)K_0(x)e^{-x^2}dx$, is annihilated by $t^4(\theta+1)^2 - t^2\theta^2(4\theta+1) + 4\theta^2(\theta-2)^2$ to $10^{-41}$, and its monodromy companion, with $I_0$ for $K_0(tx)$, grows like $e^{(t-1)^2/4}$, so every annihilator has an irregular singular point at infinity: at the level of the one function, Slater geminals stay on the ladder of Picard–Fuchs operators and Gaussian ones leave it for exponential periods (registered before computing, record docs/predictions/2026-10-22_geminal_shapes.md). **[SYMBOLIC + MEASURED]** On the full triangle, with $e^{-r^2}$ on both edges and scales $(1, 1, t)$, the Gaussian shape removes more than the elliptic piece. The minimal annihilator of $Y$ has order nine and $x$-degree 13 and splits exactly into two pieces, of orders four and five, both irregular at infinity, with formal solutions growing like $e^{t^2/4}$ and $e^{t^2/8}$; its only genuine finite singular point is $t = 2$. No piece is equivalent to $F_3$'s operator, although each keeps the K3 piece's Jordan block of size three at $t = 0$: the K3 rung of the Slater triangle is not set by the densities and the Coulomb line alone. The prediction that the K3 piece survives failed. The operator leaves $2 \times 10^{-27}$ on $Y$ at 32 digits, and an

independent evaluation of $Y$ agrees to $4 \times 10^{-14}$ (record `docs/predictions/2026-10-23_gaussian_geminal_triangle.md`, guard `tests/test_paper59_gy_gaussian.py`). Caps: one value, $\gamma = 1$, the equal-mass value, at which the Slater triangle has no elliptic piece either, so the comparison of shapes is cleanest for the K3 piece; the irreducibility of the two pieces and the absence of an $F_3$ piece rest on bounded searches; and minimality assumes that $\sqrt{\pi/2}\, e^{1/2} \operatorname{erfc}(1/\sqrt{2})$, the constant that enters the Taylor coefficients, is irrational. Guards: `tests/test_paper59_one_formula.py`, `tests/test_paper59_geminal_shapes.py`. The order-six right factor below the order-three piece leaves between 1.3 and 2.4 times the largest term of $Y$ at three values of $t$, so $Y$ needs the K3 piece, and no piece above K3 occurs (record `docs/predictions/2026-10-22_correlated_triangle_g32.md`). The singular points of $Y$'s annihilator move with $\gamma$, to $t = |1-\gamma|, 2, 1+\gamma, 2\gamma$, three of which meet at $t = 2$ for $\gamma = 1$. [**OBSERVATION**] The drop of the Coulomb ring is therefore read as a property of uniqueness, not of rings: both loops integrate out exactly, but only the Coulomb one factors into pairwise inverse distances, and only it lowers the rung.

[**OBSERVATION**] The registered decision rule did not anticipate this outcome. It allowed the Coulomb triangle to land on the chain's rung, with or without an elementary term, or to climb; it lands below. By the rule's letter $L_G$ is an order-three factor that is not equivalent to the chain's operator, which the rule called a climb, but $L_G$ factors into lower pieces, so it is not a climb in the sense the record intended. Two side readings failed. The singular points at $t = \pm 1$ were not predicted; they belong to the elliptic piece. And a Coulomb ring of three does not sit with the chain of three, as the record expected, but lower. The count of linked densities orders the trees, not the rings. The caps of Sec. VIII C apply: these are fibre statements, and the integrated value may sit lower. Further caps: for the correlated triangle, the exponential shape at five values of $\gamma$, the first the equal-mass value, with $t$ on the density carrying both geminals, and the LiH engine's cusp-correct shape at $\gamma = 3/2$ and 2 only, and a Gaussian shape at $\gamma = 1$ only; the kite operators were guessed from series and checked on their coefficients, and are now also derived by integration by parts (below), and the mechanism of the drop at $\gamma = 1$ is tested by its consequences at two values of $\gamma$, not derived; only the triangle among rings; and that the chain satisfies its own order-four $P$ rests on the Mellin contour shift. The search that excludes a K3 piece from the triangle is bounded in degree, unlike the chain's no-split proof, but the orders settle it: the factors found have orders one and two, which leaves no room for the order-three K3 factor. Guards: `tests/test_paper59_density_ring.py`.

[**PRIOR ART**] The one-center values are Remiddi's [9]: the three-electron correlation integral over Slater functions closes in logarithms and dilogarithms, and Fromm and Hill [8] treat the general case. The uniqueness relation (17) is classical; its general-dimension form, which gives the constant $\pi^3$ at $D = 3$, and the early references are collected by Preti [63]. Under the relation of Sec. VIII C to the banana ladder, the triangle's elliptic piece is the $n = 2$ rung itself. What is added here is the fibre reading: the Struve moment (20), its operator, its place below the chain, and the correlated triangle's place with it. The solid-angle form of the massive loop is Davydychev and Delbourgo's (Sec. 5.1 of [61], which recovers Nickel's result [62]); here it also rests on its numerical check. No literature search was made for the operator (21).

[**SYMBOLIC + MEASURED**] The kite operators above were guessed from Taylor series. They also follow from integration by parts, the standard reduction of Feynman integrals [10, 11]. In momentum space the kite is a two-loop integral in $D = 3$ with five lines: the two densities with index 3/2, the two geminals $(q^2 + \gamma^2)^{-2}$ and the Coulomb line $q^{-2}$. A reduction modulo a large prime, sampled at many values of $z$, gives the differential system of the master integrals, and the kite's cyclic vector in that system gives its operator. At $\gamma = 1/2$, $3/2$, 2 and 3 the operator has order six and equals the guessed one exactly. At $\gamma = 1$ the master integrals drop from nine to three and the operator is the Legendre one, so the equal-mass degeneration shows in the reduction itself. On the tetrahedron, the three-loop vacuum integral with the apex density of mass $t$, the reduction gives an operator for $Y$ of order ten at $\gamma = 3/2$, divisible on the right by the order-nine operator above. Off the symmetric slice, at density scales $(1, 2)$ and geminal exponents $(3/2, 3)$, the series route found no operator with 260 unknowns or fewer. The reduction gives one of order eleven and degree 113 in $z$, about 1300 unknowns. Its integer coefficients reach 565 bits, but the eighteen-dimensional system behind it needs only 37, so the system is rebuilt exactly from two primes and checked on a third, and the operator is built from it over $\mathbb{Q}(z)$. It agrees with the operator computed at thirteen primes, and on 300 Taylor coefficients at 280 digits it leaves $1.3 \times 10^{-280}$ of the largest term; with one coefficient changed by 1/200 it leaves $4 \times 10^{-41}$. Its genuine singular points are exactly $z = -1/4, -1, -9/4, -4, -9, -49/4, -16, -81/4$, with 0 and $\infty$: minus the squares of the two-line cuts through the two densities and through the two geminals, and of the three-line cuts that pair each density with the geminal on the opposite edge. The points $z = -25/4$ and $-25$, where a pairing of each density with its adjacent geminal would put its cuts, are ordinary, and the remaining 92 roots of the leading coefficient are apparent. Controls: the one-loop bubble in $D = 3$ against its arctan closed form, and the kite with the Coulomb line removed, whose exact Taylor series is the product of two bubbles (records `docs/predictions/2026-10-24_ibp_kite.md` and `docs/predictions/2026-10-23_price_formula_slots.md`).

[**SYMBOLIC + MEASURED**] On the tetrahedron

at the same point the reduction needs 34 masters and gives an operator for $Y/t^3$ in $x = t^2$ of order nineteen, exact over $\mathbb{Q}$ (the system rebuilt from two primes and checked on a third). On $Y$ from a 60-digit grid it leaves $5 \times 10^{-55}$ to $3 \times 10^{-43}$ of the largest term, following the quadrature error. The genuine singular points of $Y$'s minimal operator in $x > 0$ are the same eight cuts, $x = 1/4, 1, 9/4, 4, 9, 49/4, 16, 81/4$. The other roots of the leading coefficient are apparent, and at each of the eight some analytic continuation of $Y$ changes on a small loop. The change is the same to four digits in 64- and 90-digit arithmetic, and at least $10^{10}$ times the change on a loop around an ordinary point along a path at least as long. **[SYMBOLIC + MEASURED]** The order nineteen is not minimal. The reduction treats the two sectors with one geminal and no Coulomb line as different, but each is the same sunrise of the three densities times a one-line tadpole, so their masters differ by the constant ratio of the geminal scales, $\gamma_1/\gamma_2 = 1/2$. With that relation imposed the operator has order fifteen, exact over $\mathbb{Q}$, and on $Y$ it leaves $2 \times 10^{-56}$ to $4 \times 10^{-46}$ of the largest term; the same construction with the ratio $1/3$ leaves $10^{-4}$ to $2 \times 10^{-2}$. The reduction's system is block triangular in its sectors, so the pieces can be read sector by sector, and they are the predicted ones. The K3 operator of the chain of three with scales $(t; 1, 2)$ is the sunrise sector's own block, and it is a left factor. The Legendre family pulled back to the two crossed pairings, $z_1 = (4t^2+7)/(16t)$ and $z_2 = (t^2-8)/(2t)$, gives the two pieces of order two, from the sectors with one geminal and the Coulomb line; the direct pairings do not occur. The other seven pieces have order one and algebraic solutions, branched at $x = 1, 9$, at $9/4$ and at $81/4$, and come from the sectors with both geminals. Each of the three identifications is an exact map over $\mathbb{Q}(x)$, rebuilt from many primes and checked by exact division. $Y$ needs the K3 piece: the right factor without it leaves $2 \times 10^{-4}$ to $4 \times 10^{-3}$ of the largest term. It needs both elliptic pieces, since its minimal operator is genuinely singular at $x = 1/4$ and $x = 4$ and no other piece is singular there. **[SYMBOLIC + MEASURED]** The minimal operator of $Y$ has order thirteen, exact over $\mathbb{Q}$. Two of the three order-one pieces branched at $x = 9/4$ are left factors of the order-fifteen operator. The pairing of $Y$ with each of them, the bilinear concomitant, is a constant, and on $Y$ it is below $10^{-56}$ of its largest term and grows with the quadrature error; the same pairing with the K3 piece, the control, is $10^{-8}$ of its largest term and does not move at twice the step. So the right factor of order thirteen without those two pieces annihilates $Y$, leaving $8 \times 10^{-57}$ to $2 \times 10^{-46}$ of the largest term. Its only irreducible left factor is the K3 piece, once, and $Y$ needs that piece, so no smaller operator annihilates $Y$. Its five order-one pieces are branched at $x = 1, 9$ (one), at $9/4$ (one) and at $81/4$ (three). The K3 piece is irreducible: its exponents leave no room for a factor of order one or three, and its exterior square has no element of the one admissible form, so it has no factor of order two. Its monodromy preserves a symmetric form and no alternating one, with reflections at $x = 1$ and $x = 9$, and its Jordan blocks are $2 + 2$ at $x = 0$ and $3 + 1$ at infinity. Caps: the reduction is new code, checked against the guessed operators and the series; the two vanishing pairings are measured, not proved; the left-factor searches behind the order thirteen are bounded, with planted controls found; and the form, the blocks and the irreducibility are computed modulo two primes, the last by searches that are complete for its exponents. An earlier reading of the order-nineteen operator, as one piece of order twelve with none of these pieces at either end, came from a map search whose box was too small at the left end, and is withdrawn (records `docs/predictions/2026-10-23_price_formula_slots.md`, addendum 6, `docs/predictions/2026-10-25_geometric_route_y.m` and `docs/predictions/2026-10-25_tetrahedron_route_y.md`). **[PRIOR ART]** Integration by parts on this tetrahedron in momentum space is Puchalski, Kędziera and Pachucki's, for the three-electron Slater integral with integer indices [15], which they reduce to a differential equation. What is added here is the family with half-integer density lines and the geminal's squared lines, and the check against operators found independently. Guards: `tests/test_paper59_ib_kite.py` and `tests/test_paper59_y_minimal_order.py`.

**[MEASURED]** The operators are also evaluators. The kite's operator, with the kite's first six Taylor coefficients at $z = 0$ as its only input, gives the kite at all 247 nodes of the grid behind $Y$. It does so by Taylor steps along a path that passes above the three apparent singular points on the positive axis. Each step reaches half the distance to the nearest root of the leading coefficient, apparent roots included, because rounding excites solutions of the recurrence that blow up there. On one process this takes 64 s, of which the input takes 1 s; the quadrature it replaces took 7894 s on fifteen. $Y$ from the two routes agrees to $2 \times 10^{-69}$ at four values of $t$, the precision at which the grid is stored. The node error grows towards large $w$, to $3 \times 10^{-8}$ near $w = 10^{12}$, but those nodes carry so little weight that no node moves $Y$ by more than $1.4 \times 10^{-68}$. With the derivation included, the route costs about 9300 s of processor time, about a thirteenth of the upper bound on the quadrature's. The derivation is three reduction runs on six worker processes, one of which was measured at 3065 s of processor time, and a 13 s lift. Changing one operator coefficient by $1/200$ moves $Y$ by 1 to 7%. $Y$'s own operator does not pay in the same way. From the grid's data at $t = 1.7$ it reaches five other values of $t$ to $4 \times 10^{-48}$ or better, but at 4 to 12 s per value, against 0.02 s for one more grid sum, because once the kite is known the outer integral is a cheap sum. At 60 digits the structure pays where the cost is, in the inner kite. Caps: one parameter set, so a new configuration needs a new derivation, which dominates the cost; times on one machine, with the quadrature's processor time an upper bound; the com-

parison is to this paper's own grid, not to a production Slater-integral code such as Ref. [12]; and no certified error bounds, only agreement with the grid (record `docs/predictions/2026-10-26_operator_evaluator.md`, registered before computing; guard `tests/test_paper59_operator_evaluator.py`). [**MEASURED**] At chemistry precision the comparison reverses. For $Y$ to $10^{-12}$, a double-precision tanh-sinh quadrature of the same two-dimensional integral, graded at the ends and split along its ridge, takes 0.033 s of processor time on one core. The operator route takes 16 s with the operator already given, about 500 times slower, and $3 \times 10^5$ times slower with its derivation counted. It also cannot run below 60 digits: at 20 to 40 digits its Taylor walk diverges. The two routes agree on $Y$ to $7 \times 10^{-14}$. So the operator is an evaluator only where high precision is wanted; for chemistry it is a statement about structure, not a shortcut (record `docs/predictions/2026-10-28_kite_speed_test.md`, registered before timing; one parameter set, one machine). [**PRIOR ART**] Evaluating Feynman integrals by solving their differential equations in series is standard in particle physics [16–18], and Ref. [15] reduces the integer-index three-electron integral to such an equation. What is added is the measured cost on this family.

[**MEASURED**] Most of such an operator's singular points can be read off before it is derived, but not all of them. At a second set of rates, chosen blind, $(a, b, \gamma_1, \gamma_2) = (1, 3/2, 7/3, 8/3)$, the kite's operator again has eighteen masters and order eleven. The eight singular points predicted from its cuts, at minus the squared sums and differences of the rates on each cut, are all genuine and carry the predicted local exponents; the pairings of adjacent lines give none. But a ninth genuine point appears at $z = -111/1120$, with exponents $-7/2, 0, 1, \ldots, 9$, and the leading Landau equations of the kite and of its contractions have no solution there. A count of critical points of the Lee–Pomeransky likelihood, which equals the number of masters at a generic point, confirms it: 22 at generic $z$, 21 there, at two primes. The drop is in the full kite only. Its position has a closed form in the squared rates $A = a^2$, $B = b^2$, $G_1 = \gamma_1^2$, $G_2 = \gamma_2^2$:

$$z_9 = \frac{(AG_2 - BG_1)(A - B - G_1 + G_2)}{(A - G_1)(B - G_2)}. \qquad (24)$$

It is symmetric under the kite's reflection $(a, \gamma_1) \leftrightarrow (b, \gamma_2)$ and under $(a, b) \leftrightarrow (\gamma_1, \gamma_2)$. Its poles lie where an adjacent pair has equal rates, the pairings the cuts ignore. At the original rates $a\gamma_2 = b\gamma_1$, so $z_9 = 0$: the ninth point sits on $z = 0$ there, which accounts for that operator's exponent at $z = 0$ that the cuts did not explain. The formula was fitted from 41 sets of rates, 29 unknowns, and passed 18 held-out sets at two primes; it reproduces $-111/1120$ at the blind point, and on the family $(1, 2, 3/2, \gamma_2)$ the critical-point count drops from 22 to 21 exactly at $z_9$ (record `docs/predictions/2026-10-26_kite_ninth_point.md`, registered before computing). [**PRIOR ART**] The location is known. Broadhurst gives it, with $s = -z$, as the pole $s_3$ that the two-particle cuts of the kite acquire when its rung becomes massless with unequal adjacent masses (Ref. [14], Eqs. (24)–(25)); it is the massless-rung limit of the anomalous threshold of the triangle formed by the two lines of one cut and the rung, the other two lines on shell, which the kite's own leading Landau equations do not contain. [**MEASURED**] A registered test confirms that this is the mechanism here. With all five lines on shell and the loop momenta in one plane, eliminating the equations in components gives exactly the factor of Eq. (24) at three sets of rates and no other. The multipliers are all nonzero, and the first Symanzik polynomial $U$ vanishes on them, which is why a Landau test that requires $U \neq 0$ misses the point. Given a small Coulomb mass, the point does not vanish but moves: the count drops on a curve through $(z_9, 0)$, one of Broadhurst's two triangle thresholds, while the other runs to infinity as the mass goes to zero. A scan in $z$ that assumes no location finds, at the blind rates, exactly ten drop points: $z = 0$, $z_9$ and the eight cut values. Caps: the massive curve is checked modulo two primes at three masses, not over the rationals, and the local shape, the drop size on the curve and the escape direction were not run (record `docs/predictions/2026-10-27_kite_ninth_point_origin.md`, registered before computing; guard `tests/test_paper59_kite_ninth_point.py`, which also covers the blind point and Eq. (24)). What is added here is the measurement in this family: the critical-point drop, the exponent, and the complete drop locus at one set of rates. The operator's degree, 120, exceeds the prediction, and its coefficients needed a fourth prime. With $\gamma_2^2$ left free the differential system has degree 16 in it, but the kite written in the masters has degree 26, so the one-parameter family exceeded its registered sample cap and was not built (record `docs/predictions/2026-10-26_parametric_kite_operator.md`, registered before computing). [**PRIOR ART**] Reading singular points from cuts is standard Landau and threshold analysis.

[**MEASURED**] In the molecular calculation the triangle class is a modest share of the cost. In the explicitly correlated LiH calculation of Paper 19 at its best truncation, the class that $Y$ models takes 21% of a one-process run, third of its four large pieces, behind the electron–nucleus and kinetic blocks. The part an operator evaluator would replace, the factorization of the three-body kernels, is 3% of the run; the rest of the class is about 2.4 million Coulomb mode solves. So a free triangle class would speed the run by at most 1.27 times, and a free kernel build by 1.03 times. Caps: one run per truncation on one machine; a reduced truncation is no proxy, giving 48% (record `docs/predictions/2026-10-26_lih_triangle_share.md`, registered before computing).

[**MEASURED**] Whether the one-mass slice could

be tabulated, as Gaussian codes tabulate the Boys function, depends on how many master integrals it needs and in how many variables. Counting critical points, as for the kite, the slice needs four masters, and the index range the production code uses needs a fifth, an antiderivative in $D$ from a pole at $x = 0$; a scaling identity leaves two variables. By the registered gate this is borderline: a table is not ruled out, but its speed against a production code [12] is untested. The physical fibre, with two offsets, needs six masters in three variables. Caps: the scaling identity missed its registered precision, $2.7 \times 10^{-25}$ against $10^{-25}$, and an unregistered rerun agrees to $1.6 \times 10^{-37}$; these are counts of masters, not the cost of a table (record `docs/predictions/2026-10-27_master_count_tabulation.md` registered before computing; guard `tests/test_paper59_master_count.py`).

[**MEASURED**] Production explicitly correlated codes do not use a single Gaussian geminal. For many integrals they replace $e^{-\gamma r}$ by a least-squares fit, six Gaussians by default. The off-slice triangle above, known to about fifty digits, measures that replacement. With the fit in the plain norm $\int_0^\infty r^2(\cdot)^2\,dr$, which at three and six terms gives the STO-3G and STO-6G exponents, the six-term fit makes $Y$ too small by $3.6 \times 10^{-5}$ at $t = 1/2$ down to $2.4 \times 10^{-6}$ at $t = 4$. The error falls like $e^{-b\sqrt{n}}$ in the number of terms $n$, as the cusp at $r = 0$ predicts. With an extra weight $e^{-\gamma^2 r^2}$ the fit loses its tail, and the six-term error is $5.4 \times 10^{-2}$ to $1.7 \times 10^{-2}$. So on this integral the error is set by how far out the fit is trusted, not by the cusp. Caps: the production weight function is not reproduced, so these two weights do not price a production fit; one configuration, with model densities (record `docs/predictions/2026-10-25_gaussian_fit_geminal.md` registered before computing). [**MEASURED**] A fit that a production code ships, the six-term fit written into Psi4's source [19], is the best of the three at the cusp and the worst in the tail. It is 1.8% low at $r = 0$, against 6% for the plain fit, and half the true value at $\gamma r = 4$. On the same $Y$ it gives an error of $-13\%$ at $t = 1/2$ down to $-5.5\%$ at $t = 4$, more than three thousand times the plain fit's. The three fits are ordered by their tails, not by their cusps. Caps: $Y$ weights the geminal over its whole range, while a correlation energy is dominated by short range, so this prices the fit on this integral and not on an F12 energy; the fit's weight and its literature source are not verified, nor which integrals the code uses it for (record `docs/predictions/2026-10-26_production_geminal_fit.md` registered before computing; guard `tests/test_paper59_production_geminal_fit.py`).

## IX. SCOPE, AND THE OPEN FRONTIER

*Scope.* The demonstration is for $1s$ orbitals at a common exponent. Higher $(n, \ell, m)$ multiply the integrand of Eq. (3) by solid-harmonic polynomials in $\boldsymbol{k}$ and shift the exponents; these change the weights and arguments but not the genus of the curve (5), which is set by the two distinct scales. The result is a statement about the transcendence *class* of the three-center integral, established on its simplest genuine instance. It does not by itself deliver a polyatomic energy: a molecular Hamiltonian requires the full three-center tensor, and a value that is part closed-form and part numerical is neither more accurate (the Gaussian fits are already exact to fit quality) nor decidable. What the reduction buys is an accurate, fast, geometry-native evaluator for the three-center integral and—the point of this paper—a precise diagnosis of why it is hard.

*The wall is closed-form, not computational.* [**MEASURED**] The infinite series that the standard verdict falls back on is, concretely, the expansion of both densities in multipoles about the shared center (the Barnett–Coulson / Löwdin $\alpha$-function route [7]; a recent analytic treatment of the same integrals, for linear geometries, is Ref. [6]; both are the numerical treatments named in Sec. VI). Each channel is a product of two one-pair factors, and the third center enters only through the relative orientation of the two pairs. On the water geometry ($R_{\mathrm{OH}} = 1.809$, $104.5^\circ$) with O $1s/2s/2p$ and H $1s$ Slater orbitals, seven three-center integrals of both shared-O and shared-H type converge to $\sim 10^{-9}$ by $L \approx 12$. Gaussian-fit sequences converge onto every value; for the O $1s$ core the Gaussians approach it from below and are still $6 \times 10^{-8}$ short at 18 Gaussians, while the channel sum is stable to thirteen digits (`tests/test_pairwise_general.py`, `tests/test_pairwise_three_center.py`). [**MEASURED**] The channel sum is the Legendre series of the integral in $\cos\gamma$, $\gamma$ the bond angle, so the decay of its terms locates the nearest singular point of the integral as a function of the angle. In every case tested the terms keep one sign, so that point is not the linear geometry. At water's equal bond lengths they fall like a power, about $\ell^{-8}$, which puts it at $\gamma = 0$, where the two hydrogens would merge; at unequal lengths $D_1 \neq D_2$ they fall geometrically, at a rate within 3% of $D_</D_>$, the rate of $1/|Y - Z|$, whose only singular point is the two outer centers merging. Bent water is an ordinary point. Caps: the registered tolerance for that rate was 1% and it was missed in all four cases; $1s$ orbitals with two exponent sets only (record `docs/predictions/2026-10-27_water_angle_singularities.md`, registered before computing; guard `tests/test_paper59_water_bond_angle.py`). [**SYMBOLIC + MEASURED**] The singular term itself is derived. Only the two hydrogen cusps, meeting through $1/r_{12}$, make the integral non-analytic in the angle, and the lowest such term is $C\,|Y - Z|^7$ with $C = \zeta_{\mathrm{H}}^5 \zeta_{\mathrm{X}}^3 e^{-\zeta_{\mathrm{X}}(D_1+D_2)}/630$. The same local computation, run on the two-center Coulomb integral of two $1s$ densities, reproduces the first two odd terms of its closed form, $(2/315)R^7 + (16/14175)R^9$, exactly. The term factors into one number per hydrogen. Call $g_Y$

the coefficient of $|\mathbf{r}-Y|$ in the density $\chi_X\chi_Y$ at $Y$, its contact; Kato's cusp rule sets it to $-\zeta_{\rm H}\chi_X(Y)\chi_Y(Y)$. Then the term is $(\pi^2/630)\,g_Y g_Z\,|Y-Z|^7$, and $\pi^2/630$ is the same for any two densities with a point cusp. For Slater orbitals the $\pi^2$, a Fourier-measure factor, cancels against the normalizations, so $C$ is $\pi$-free. This is the Coulomb form of Tan's contact relations, in which one short-distance number per cusp, times a universal function, fixes a slow tail [24]. The small-$R$ series of two-center Coulomb integrals, with a non-analytic family of terms beside the even one, is Silverstone's [26]. For the one-electron analogue, the three-center nuclear-attraction integral, Bouferguene and Jones derived the large-$\ell$ terms of the same multipole series: geometric in the ratio of the two distances, times a power of $\ell$ set by the orbital, and only logarithmically convergent when the distances are equal [25]. The two-electron cusp-cusp term, its size, and its reading as the bond-angle singularity were not found in a literature search (record `docs/predictions/2026-10-28_water_landau_gaussian_fresh.md`); the full text of Ref. [26] was not read, so whether it gives the two-center $R^7$ coefficient in this form is unchecked. The term fixes the power: $\ell^{-8}$ at equal lengths, and $\ell^{-4}$ times the geometric rate at unequal ones. At water's equal lengths the first two orders, $\alpha_7(1-x)^{7/2}+\alpha_9(1-x)^{9/2}$, both in closed form from the two exponents and the bond length, reproduce the coefficients to $10^{-4}$ at $\ell=60$ with no fitted number, and what is left falls like $\ell^{-4}$, the order of the first term left out. The rate test above missed because its fit left the power free: on a function with exactly this singularity that fit is 5% off, and with the derived power the four unequal-length rates come within 0.03% (reported only; that run was voided by one of its own controls; records `docs/predictions/2026-10-27_water_cusp_singularity.md` and `docs/predictions/2026-10-27_water_cusp_next_term.md`, each registered before computing). **[SYMBOLIC + MEASURED]** At unequal lengths the singular point moves to $x^*=(D_1^2+D_2^2)/(2D_1D_2)>1$, where $|Y-Z|^2=0$ at a complex angle, and every gradient of the two cusp functions enters at the same order there, so the endpoint value of $C$ does not carry over. Summed, those orders are the moments of $24su$ on the simplex $s,u\ge0$, $s+u\le1$: the term is $\Phi\,(2D_1D_2)^{7/2}(x^*-x)^{7/2}$ with $\Phi=(\pi^2/630)\,24\iint su\,G_Y(-sw)\,G_Z(uw)\,ds\,du$, $w=Y-Z$, the two cusp functions averaged along the segment between the hydrogens. This is the collinear pinch of the Landau equations for $|\mathbf{a}|$, $|\mathbf{b}|$ and $1/|w+\mathbf{a}-\mathbf{b}|$. For $1s$ functions $\Phi$ is a two-dimensional integral of $e^{-\zeta_{\rm X}(R_1(s)+R_2(u))}$ with $R_1(s)^2=(1-s)D_1^2+sD_2^2$, and it equals $C$ at $D_1=D_2$. Extracted from the coefficients with the subleading powers as fitted nuisances, the leading size matches $\Phi$ to $3.2\times10^{-5}$ in four cases registered before computing, with both lengths and the hydrogen exponent varied and nothing fitted in the size; the endpoint value is off by 6.5 to 25% (record `docs/predictions/2026-10-28_water_landau_gaussian_fresh.md`, a first registration stopped on a control bound set without measuring, record `docs/predictions/2026-10-28_water_landau_unequal.md`). The pinch mechanism is standard, and taking the coefficient at one point is the rule for a simple pinch; here the pinch points fill a segment. Its use for this term, and the segment-average size, were not found in a literature search (same record). Caps: $1s$ functions, oxygen exponents 3 and 4. **[SYMBOLIC + MEASURED]** With zero contact the $\ell^{-8}$ term is absent. For a hydrogen function whose first-order cusp cancels, the derived next term, $(1-x)^{9/2}$ with one such function and $(1-x)^{11/2}$ with two, reproduces the coefficients with nothing fitted, with the sign the derivation gives: to 0.5–1.9% at $\ell=35$ on four cases registered before computing (two cusp-free pairs, two oxygen exponents, two bond lengths; record `docs/predictions/2026-10-27_water_zero_contact_fresh.md`). A first registration stopped on a sign clause it stated wrongly; its size and remainder clauses held (record `docs/predictions/2026-10-27_water_zero_contact.md`). Caps: symmetric geometry, $1s$ on the oxygen, cusp-free pairs of $1s$ functions only. **[MEASURED]** Gaussian functions have no cusp, so the contact reading says their bond-angle series has no power-law tail. For one $s$ Gaussian per center the integral is a Boys function, linear in $\cos\gamma$, and its coefficients fall like $T^\ell/(2\ell+1)!!$. STO-3G and STO-6G contractions follow the Slater $\ell^{-8}$ coefficients up to a crossover $\ell^*\approx3.6$–$3.7\,D\sqrt{\alpha_{\max}}$, $\alpha_{\max}$ the tightest hydrogen exponent, and fall away past it. A model that keeps only the cusp-cusp term placed $\ell^*$ to within one in four cases registered before computing, and $\ell^*$ moves with $\sqrt{\alpha_{\max}}$ (record `docs/predictions/2026-10-28_water_landau_gaussian_fresh.md`, a first registration stopped on a control whose precision it did not set, record `docs/predictions/2026-10-28_water_gaussian_contact.md`). That a Gaussian fit copies a cusp only down to about $1/\sqrt{\alpha_{\max}}$ is basis-set lore; this is a check of the contact reading, not a new result. Caps: symmetric geometry, $s$ functions, two published contractions. **[SYMBOLIC + MEASURED]** The bond-angle function obeys a linear differential equation in $x$, since the integrand is holonomic in the positions; we looked for it. Each Legendre coefficient is a finite sum of rational numbers times fixed constants, built from the three-term rules of $i_\ell$ and $k_\ell$ in $\ell$ and exact integration by parts: six constants for the one-electron three-center integral $\int\phi_{\rm X}\phi_{\rm Y}/|\mathbf{r}-Z|$ (four exponentials and $e^{\mp\zeta_{\rm H}D_1}E_1((\zeta_{\rm X}+\zeta_{\rm H})D_2)$), and 31 for $(XY|XZ)$, all exponentials times 1, $E_1$ or a logarithm, so each channel stays in the two-center class. With the coefficients exact, the equation is found by linear algebra modulo primes, where an approximate fit cannot imitate it; from floating-point coefficients at 110 to 200 digits, approximate low-order fits matched even a known control to the data's precision and hid it. The one-electron integral obeys an operator of order 8 with polynomial coefficients of degree 31, the same in two geometries and modulo two primes; an independent derivation from the $\ell$-recurrences alone, with no fit

to the coefficients, gives an order-9 operator that the order-8 one divides exactly. Its only finite singular point with a non-integer exponent is $x^*$, with exponents 3/2 and 5/2 besides integers: Poisson's equation turns the density's $|\mathbf{r} - Y|$ cusp term into a $|Z - Y|^3$ term of the potential. The points $x = \pm 1$ carry repeated integer exponents and one further rational point an exponent $-2$; the integral is analytic at both, so these belong to other solutions. Infinity is irregular. For $(XY|XZ)$ no operator exists with fewer than about 1970 unknowns at any order up to 40 (2001 coefficients, two geometries, two primes); the constant components alone carry seven distinct operators of orders 6 to 13 (records `docs/predictions/2026-10-28_nuclear_attraction_angle_operator.md`, `docs/predictions/2026-10-28_bottom_up_angle_operator.md`, `docs/predictions/2026-10-28_two_electron_angle_operator.md`, and `docs/predictions/2026-10-28_angle_operator.md`, each registered before computing). Built up from the $\ell$-recurrences in the same way, with no fit, the $(XY|XZ)$ coefficients obey a recurrence in $\ell$ of order 41 and degree 926 (one geometry, two primes; it annihilates the exact coefficients up to $\ell = 400$). Among its rational characteristic roots, only those at $x^*$ carry exponents of a branch point. Evaluated to $\ell = 300$ with more than 550 digits, the exact coefficients decay as $\rho^\ell \ell^{-4}$, $\rho = D_</D_>$, so the slowest bond-angle term is $(x^* - x)^{7/2}$, the cusp-cusp term $|Y - Z|^7$ above. The recurrence's other solutions, among them $\ell^3$ and $\ell^1$ at $x^*$, carry zero weight (records `docs/predictions/2026-10-28_two_electron_bottom_up.md` and `docs/predictions/2026-10-29_two_electron_large_l.md`, each registered before computing). Guessing an operator from many exact terms is standard [45, 46]; an equation in the bond angle for a three-center integral was not found in a literature search. Caps: $1s$ functions; the constants are assumed linearly independent over $\mathbb{Q}$. The elliptic content of Eq. (5) is spread over the infinite channel sum. The radial core is a period of an irreducible connection (Sec. V, **[SYMBOLIC + MEASURED]**), so it has no closed form through any factorization; that does not bar an accurate evaluation, and whether the integrated three-center integral has a finite closed form is the open problem below.

*The open problem.* As shown in Sec. V, the elliptic-dilogarithm route is obstructed: $N(D)$ is not an elliptic dilogarithm of elementary or period-level data but a period of the rank-four irregular connection (9). We settle the differential-Galois *reducibility*: the fourth-order operator is irreducible over $\mathbb{C}(D)$—hence over $\mathbb{Q}(\rho)(D)$—because it is the Fourier–Laplace transform of an irreducible connection, confirmed concretely by its exact-integer monodromy (Sec. V), so $N(D)$ admits no closed form through any factorization of the operator. The one remaining question is whether the irreducible connection nonetheless carries a *transcendental* closed form, which we do not settle. As a function of the bond angle the integrated object is large: its operator in $x$ has more than about 1970 unknowns at every order up to 40, where the one-electron analogue needs order 8, though its Legendre coefficients obey a recurrence in $\ell$ of order 41 derived from the $\ell$-recurrences (Sec. IX). The companion (Paper 61) localizes it: the object is an Eisenstein / complex-multiplication Bessel-moment period—not a cusp-form $L$-value and not an elliptic dilogarithm—whose master family obeys the quadratic period relations $B = \pi\,\Omega$, $\Omega$ the canonical symplectic intersection form now identified in closed form (developed in the companion, Paper 61), preserved by the integral monodromy; the open piece is the explicit finite reduction of the observable itself to $\pi$ and the CM-fibre $\Gamma$-values. (This tempers a literature-scout expectation that the object would sit at the tractable elliptic-dilogarithm level because it has one operator propagator factors; that pattern-match from the sunrise is overridden by the in-module computation of Sec. V.) Because it continues the Averys' momentum-space Sturmian program and requires elliptic-Feynman-integral technique one level beyond the sunrise, it is the natural place for collaboration.

## X. CONCLUSION

The three-center electron-repulsion integral is not merely a harder version of the two-center one; it is an object of higher genus. Two centers close at genus zero over $\{E_1, \ln, \gamma\}$; three centers, read in momentum space where the coordinate obstruction dissolves, reduce to a two-scale Bessel moment on an elliptic curve, an elliptic deformation of the Bessel $K_0$ with an explicit fourth-order Picard–Fuchs equation and a period that is a complete elliptic integral. The seventy-year difficulty of polyatomic Slater integrals is, at the level of transcendence, a genus jump; and a classical molecular integral turns out to be an instance of the modern theory of elliptic Feynman integrals—a bridge that, to our knowledge, had not been drawn. With a Slater geminal the two-center master integral already reaches the same curve: its published fourth-order operator is the three-center one rescaled, and only symmetric exponent patterns collapse. The finite closed form remains open—and, we have shown, past the reach of the sunrise's elliptic-dilogarithm mechanism, whose inhomogeneity is absent here: the object is a period of a rank-four irregular connection which we prove irreducible—it is the Fourier–Laplace transform of an irreducible connection, and its monodromy admits no invariant coordinate subspace—so it does not close by any factorization. Whether it closes at the irreducible level—as an Eisenstein/CM Bessel-moment period, a finite reduction to $\pi$ and the CM-fibre $\Gamma$-values (companion, Paper 61)—is the one frontier that remains, posed here as it deserves to be. Two pre-registered extensions (Sec. VIII) probe the rung variable: a fourth center adds nothing, consistent with the class counting the distinct two-center densities a kernel couples and not the nuclei, while a second Coulomb kernel at one electron, over two-center densities, climbs to a K3 whose period is elliptic squared. A third extension shows the rung of a tree's

holomorphic period is set by how many densities it links: four linked densities give the first period not built from elliptic ones, five give no new type, and the integrals of a molecular Hamiltonian, which link two, stay elliptic at the fibre for any molecule; the operators for two to five densities are known, tied to the banana ladder by a Hadamard product, and what is new is the route from a molecular integral. A fourth shows that the count orders trees, not rings: a Coulomb triangle of three densities sits on the elliptic rung at the fibre, below the chain of three, while the correlated triangle with exponential geminals sits with the chain at $\gamma = 1$; we read the difference as uniqueness, which factors the Coulomb loop and not the geminal one. Off the symmetric slice the correlated triangle carries the K3 piece of the chain of three, an elliptic piece for each crossed pairing of a density with a geminal, and five algebraic pieces; its minimal operator has order thirteen. The operators also evaluate: the kite's, from six Taylor coefficients, rebuilds the grid behind $Y$ to the grid's own precision in about a thirteenth of the upper bound on the quadrature's processor time, derivation included, while $Y$'s own operator is slower than one more grid sum. That gain is a property of 60-digit arithmetic: at $10^{-12}$ a double-precision quadrature is about 500 times faster than the kite's operator, so for chemistry the operator is structure, not a shortcut. At a second set of rates the cuts predict eight of the kite's nine singular points; the ninth has a measured closed form in the squared rates, a known triangle threshold of the massless-rung kite (Broadhurst), confirmed here as an on-shell pinch that moves, not vanishes, at a small Coulomb mass, and it sits on $z = 0$ at the original rates. In the LiH calculation the triangle class is a fifth of the run and its kernel build three percent. A six-Gaussian geminal fit that a production code ships misses $Y$ by five to thirteen percent, through its tail and not its cusp.


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